Crab: I have never seen that illustration before. Why don't you pass it 'round?

(The Tortoise passes the book around. Each of the foursome looks at it in a characteristic way-this one from afar, that one from close up, everyone tipping his head this way and that in puzzlement. Finally it has made the rounds, and returns to the Tortoise, who peers at it rather intently.)

Achilles: Well, I guess the prelude is just about over. I wonder if, as I listen to this fugue, I will gain any more insight into the question, "What is the right way to listen to a fugue: as a whole, or as the sum of its parts?"

TTortoise: Listen carefully, and you will!

(The prelude ends. There is a moment of silence; and ...

[ATTACCA]

CHAPTER X

Levels of Description, and Computer Systems

Levels of Description

GODEL'S STRING G, and a Bach fugue: they both have the property that they can be understood on different levels. We are all familiar with this kind of thing; and yet in some cases it confuses us, while in others w handle it without any difficulty at all. For example, we all know that w human beings are composed of an enormous number of cells (around twenty-five trillion), and therefore that everything we do could in principle be described in terms of cells. Or it could even be described on the level c molecules. Most of us accept this in a rather matter-of-fact way; we go t the doctor, who looks at us on lower levels than we think of ourselves. W read about DNA and "genetic engineering" and sip our coffee. We seem t have reconciled these two inconceivably different pictures of ourselves simply by disconnecting them from each other. We have almost no way t relate a microscopic description of ourselves to that which we feel ourselves to be, and hence it is possible to store separate representations of ourselves in quite separate "compartments" of our minds. Seldom do we have to fir back and forth between these two concepts of ourselves, wondering "How can these two totally different things be the same me?"

Or take a sequence of images on a television screen which show Shirley MacLaine laughing. When we watch that sequence, we know that we are actually looking not at a woman, but at sets of flickering dots on a flat surface. We know it, but it is the furthest thing from our mind. We have these two wildly opposing representations of what is on the screen, but that does not confuse us. We can just shut one out, and pay attention to th other-which is what all of us do. Which one is "more real"? It depends o; whether you're a human, a dog, a computer, or a television set.

Chunking and Chess Skill

One of the major problems of Artificial Intelligence research is to figure out how to bridge the gap between these two descriptions; how to construe a system which can accept one level of description, and produce the other One way in which this gap enters Artificial Intelligence is well illustrated b the progress in knowledge about how to program a computer to play goof chess. It used to be thought in the 1950's and on into the 1960's-that the

trick to making a machine play well was to make the machine look further ahead into the branching network of possible sequences of play than any chess master can. However, as this goal gradually became attained, the level of computer chess did not have any sudden spurt, and surpass human experts. In fact, a human expert can quite soundly and confidently trounce the best chess programs of this day.

The reason for this had actually been in print for many years. In the 1940's, the Dutch psychologist Adriaan de Groot made studies of how chess novices and chess masters perceive a chess situation. Put in their starkest terms, his results imply that chess masters perceive the distribution of pieces in *chunks*. There is a higher-level description of the board than the straightforward "white pawn on K5, black rook on Q6" type of description, and the master somehow produces such a mental image of the board. This was proven by the high speed with which a master could reproduce an actual position taken from a game, compared with the novice's plodding reconstruction of the position, after both of them had had five-second glances at the board. Highly revealing was the fact that masters' mistakes involved placing whole *groups* of pieces in the wrong place, which left the game strategically almost the same, but to a novice's eyes, not at all the same. The clincher was to do the same experiment but with pieces randomly assigned to the squares on the board, instead of copied from actual games. The masters were found to be simply no better than the novices in reconstructing such random boards.

The conclusion is that in normal chess play, certain types of situation recur-certain patterns-and it is to those high-level patterns that the master is sensitive. He thinks *on a different level* from the novice; his set of concepts is different. Nearly everyone is surprised to find out that in actual play, a master rarely looks ahead any further than a novice does-and moreover, a master usually examines only a handful of possible moves! The trick is that his mode of perceiving the board is like a filter: he literally *does not see bad moves* when he looks at a chess situation-no more than chess amateurs see illegal moves when they look at a chess situation. Anyone who has played even a little chess has organized his perception so that diagonal rook-moves, forward captures by pawns, and so forth, are never brought to mind. Similarly, master-level players have built up higher levels of organization in the way they see the board; consequently, to them, bad moves are as unlikely to come to mind as illegal moves are, to most people. This might be called *implicit pruning* of the giant branching tree of possibilities. By contrast, *explicit pruning* would involve thinking of a move, and after superficial examination, deciding not to pursue examining it any further.

The distinction can apply just as well to other intellectual activities -- for instance, doing mathematics. A gifted mathematician doesn't usually think up and try out all sorts of false pathways to the desired theorem, as less gifted people might do; rather, he just "smells" the promising paths, and takes them immediately.

Computer chess programs which rely on looking ahead have not been taught to think on a higher level; the strategy has just been to use brute

force look-ahead, hoping to crush all types of opposition. But it h worked. Perhaps someday, a look-ahead program with enough brute ,gill indeed overcome the best human players-but that will be a intellectual gain, compared to the revelation that intelligence de crucially on the ability to create high-level descriptions of complex such as chess boards, television screens, printed pages, or painting

Similar Levels

usually, we are not required to hold more than one level of understanding of a situation in our minds at once. Moreover, the different descriptions a single system are usually so conceptually distant from each other tl was mentioned earlier, there is no problem in maintaining them both are just maintained in separate mental compartments. What is confusing though, is when a single system admits of two or more descriptions different levels which nevertheless *resemble* each other in some way. we find it hard to avoid mixing levels when we think about the system can easily get totally lost.

Undoubtedly this happens when we think about our psychology-for instance, when we try to understand people's motivations: for various actions. There are many levels in the human m structure-certainly it is a system which we do not understand very we But there are hundreds of rival theories which tell why people act the way they do, each theory based on some underlying assumptions about he down in this set of levels various kinds of psychological "forces" are f(Since at this time we use pretty much the same kind of language f mental levels, this makes for much level-mixing and most certain] hundreds of wrong theories. For instance, we talk of "drives"-for se power, for fame, for love, etc., etc.-without knowing where these drives come from in the human mental structure. Without belaboring the pc simply wish to say that our confusion about who we are is certainly r(to the fact that we consist of a large set of levels, and we use overlapping language to describe ourselves on all of those levels.

Computer Systems

There is another place where many levels of description coexist for a system, and where all the levels are conceptually quite close to one an(I am referring to computer systems. When a computer program is ping, it can be viewed on a number of levels. On each level, the description is given in the language of computer science, which makes all the de descriptions similar in some ways to each other-yet there are extremely imp(differences between the views one gets on the different levels. At the 1 level, the description can be so complicated that it is like the dot-description of a television picture. For some purposes, however, this is by far the important view. At the highest level, the description is greatly *chunked* and

takes on a completely different feel, despite the fact that many of the same concepts appear on the lowest and highest levels. The chunks on the high-level description are like the chess expert's chunks, and like the chunked description of the image on the screen: they summarize in capsule form a number of things which on lower levels are seen as separate. (See Fig. 57.) Now before things become too abstract, let us pass on to the

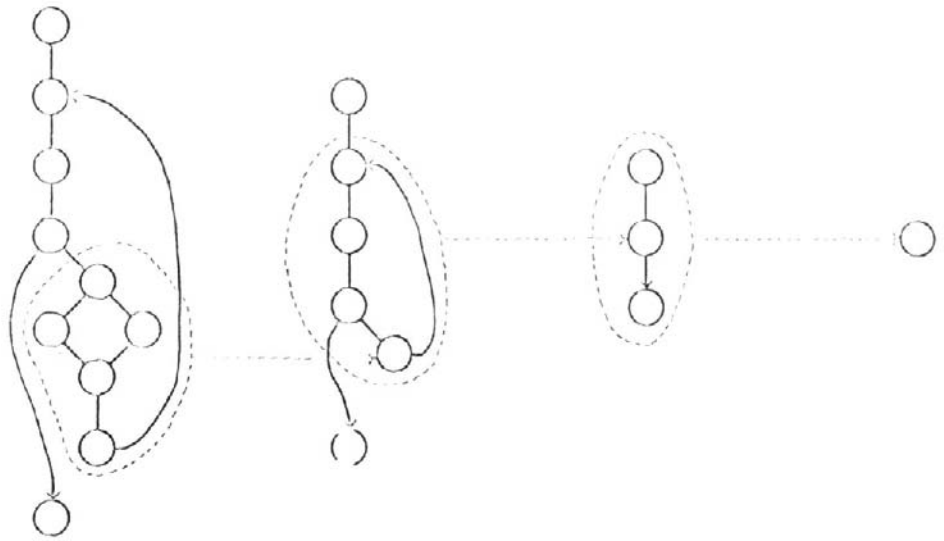
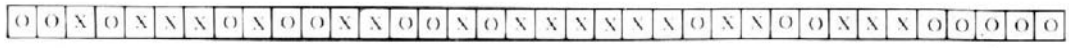


FIGURE 57. The idea of "chunking": a group of items is reperceived as a single "chunk". The chunk's boundary is a little like a cell membrane or a national border: it establishes a separate identity for the cluster within. According to context, one may wish to ignore the chunk's internal structure or to take it into account.

concrete facts about computers, beginning with a very quick skim of what a computer system is like on the lowest level. The lowest level? Well, not really, for I am not going to talk about elementary particles-but it is the lowest level which we wish to think about.

At the conceptual rock-bottom of a computer, we find a memory, a central processing unit (CPU), and some input-output (I/O) devices. Let us first describe the memory. It is divided up into distinct physical pieces, called words. For the sake of concreteness, let us say there are 65,536 words of memory (a typical number, being 2 to the 16th power). A word is further divided into what we shall consider the atoms of computer science-bits. The number of bits in a typical word might be around thirty-six. Physically, a bit is just a magnetic "switch" that can be in either of two positions.



--- a word of 36 bits --

you could call the two positions "up" and "down", or "x" and "o", o and "0" ... The third is the usual convention. It is perfectly fine, but i the possibly misleading effect of making people think that a comp deep down, is storing numbers. This is not true. A set of thirty-six bits not have to be thought of as a number any more than two bits has i thought of as the price of an ice cream cone. Just as money can do va things depending on how you use it, so a word in memory can serve r functions. Sometimes, to be sure, those thirty-six bits will indeed repn a number in binary notation. Other times, they may represent thin dots on a television screen. And other times, they may represent a letters of text. How a word in memory is to be thought of depends eni on the role that this word plays in the program which uses it. It ma course, play more than one role-like a note in a canon.

Instructions and Data

There is one interpretation of a word which I haven't yet mentioned, that is as an instruction. The words of memory contain not only data t acted on, but also the program to act on the data. There exists a lin repertoire of operations which can be carried out by the central proce5 unit-the CPU-and part of a word, usually its first several bits-is it pretable as the name of the instruction-type which is to be carried What do the rest of the bits in a word-interpreted-as-instruction stand Most often, they tell which other words in memory are to be acted upoi other words, the remaining bits constitute a pointer to some other wor(words) in memory. Every word in memory has a distinct location, li house on a street; and its location is called its address. Memory may have "street", or many "streets"-they are called "pages". So a given wo addressed by its page number (if memory is paged) together wit position within the page. Hence the "pointer" part of an instruction i numerical address of some word(s) in memory. There are no restric on the pointer, so an instruction may even "point" at itself, so that whet executed, it causes a change in itself to be made.

How does the computer know what instruction to execute at any € time? This is kept track of in the CPU. The CPU has a special pointer w points at (i.e., stores the address of) the next word which is to be inter ed as an instruction. The CPU fetches that word from memory, and c it electronically into a special word belonging to the CPU itself. (Wor the CPU are usually not called "words", but rather, registers.) Then the executes that instruction. Now the instruction may call for any of a number of types of operations to be carried out. Typical ones include:

ADD the word pointed to in the instruction, to a register.
(In this case, the word pointed to is obviously interpreted as number.)

PRINT the word pointed to in the instruction, as letters.

(In this case, the word is obviously interpreted *not* as a number, but as a string of letters.)

JUMP to the word pointed to in the instruction.

(In this case, the **CPU** is being told to interpret that particular word as its next instruction.)

Unless the instruction explicitly dictates otherwise, the **CPU** will pick up the very next word and interpret it as an instruction. In other words, the **CPU** assumes that it should move down the "street" sequentially, like a mailman, interpreting word after word as an instruction. But this sequential order can be broken by such instructions as the **JUMP** instruction, and others.

Machine Language *vs*. Assembly language

This is a very brief sketch of *machine language*. In this language, the types of operations which exist constitute a finite repertoire which cannot be extended. Thus all programs, no matter how large and complex, must be made out of compounds of these types of instructions. Looking at a program written in machine language is vaguely comparable to looking at a **DNA** molecule atom by atom. If you glance back to Fig. 41, showing the nucleotide sequence of a **DNA** molecule--and then if you consider that each nucleotide contains two dozen atoms or so-and if you imagine trying to write the **DNA**, atom by atom, for a small virus (not to mention a human being!)-then you will get a feeling for what it is like to write a complex program in machine language, and what it is like to try to grasp what is going on in a program if you have access only to its machine language description. ,

It must be mentioned, however, that computer programming was originally done on an even lower level, if possible, than that of machine language--namely, connecting wires to each other, so that the proper operations were "hard-wired" in. This is so amazingly primitive by modern standards that it is painful even to' imagine. Yet undoubtedly the people who first did it experienced as much exhilaration as the pioneers of modern computers ever do .. .

We now wish to move to a higher level of the hierarchy of levels of description of programs. This is the *assembly language* level. There is not a gigantic spread between assembly language and machine language; indeed, the step is rather gentle. In essence, there is a one-to-one correspondence between assembly language instructions and machine language instructions. The idea of assembly language is to "chunk" the individual machine language instructions, so that instead of writing the sequence of bits "010111000" when you want an instruction which adds one number to another, you simply write **ADD**, and then instead of giving the address in binary representation, you can refer to the word in memory by a *name*.

Therefore, a program in assembly language is very much like a machine language program made legible to humans. You might compare the machine language version of a program to a **TNT**-derivation done in the obscure Gödel-numbered notation, and the assembly language version to the isomorphic **TNT**-derivation, done in the original **TNT**-notation, which is much easier to understand. Or, going back to the **DNA** image, we can liken the difference between machine language and assembly language to the difference between painfully specifying each nucleotide, atom by atom, and specifying a nucleotide by simply giving its *name* (i.e., **'A', 'G', 'C', or 'T'**). There is a tremendous saving of labor in this very simple "chunking" operation, although conceptually not much has been changed.

Programs That Translate Programs

Perhaps the central point about assembly language is not its differences from machine language, which are not that enormous, but just the key idea that programs could be written on a different level *at all*! Just think about it: the hardware is built to "understand" machine language programs-sequences of bits-but not letters and decimal numbers. What happens when hardware is fed a program in assembly language% It is as if you tried to get a cell to accept a piece of paper with the nucleotide sequence written out in letters of the alphabet, instead of in chemicals. What can a cell do with a piece of paper? What can a computer do with an assembly language program?

And here is the vital point: someone can write, in machine language, a *translation program*. This program, called an *assembler*, accepts mnemonic instruction names, decimal numbers, and other convenient abbreviations which a programmer can remember easily, and carries out the conversion into the monotonous but critical bit-sequences. After the assembly language program has been *assembled* (i.e., translated), it is *run*-or rather, its machine language equivalent is run. But this is a matter of terminology. Which level program is running? You can never go wrong if you say that the machine language program is running, for hardware is always involved when any program runs-but it is also quite reasonable to think of the running program in terms of assembly language. For instance, you might very well say, "Right now, the **CPU** is executing a **JUMP** instruction", instead of saying, "Right now, the **CPU** is executing a ' 1 11010000' instruction". A pianist who plays the notes **G-E-B E-G-B** is also playing an arpeggio in the chord of E minor. There is no reason to be reluctant about describing things from a higher-level point of view. So one can think of the assembly language program running concurrently with the machine language program. We have two modes of describing what the **CPU** is doing.

Higher-Level Languages, Compilers, and Interpreters

The next level of the hierarchy carries much further the extremely powerful idea of using the computer itself to translate programs from a high level into lower levels. After people had programmed in assembly language for a number of years, in the early 1950's, they realized that there were a number of characteristic structures which kept reappearing in program after program. There seemed to be, just as in chess, certain fundamental patterns which cropped up naturally when human beings tried to formulate algorithms--exact descriptions of processes they wanted carried out. In other words, algorithms seemed to have certain higher-level components, in terms of which they could be much more easily and esthetically specified than in the very restricted machine language, or assembly language. Typically, a high-level algorithm component consists not of one or two machine language instructions, but of a whole collection of them, not necessarily all contiguous in memory. Such a component could be represented in a higher-level language by a single item-a chunk.

Aside from standard chunks-the newly discovered components out of which all algorithms can be built-people realized that almost all programs contain even larger chunks-superchunks, so to speak. These superchunks differ from program to program, depending on the kinds of high-level tasks the j program is supposed to carry out. We discussed superchunks in Chapter V, calling them by their usual names: "subroutines" and "procedures". It was clear that a most powerful addition to any programming language would be the ability to define new higher-level entities in terms of previously known ones, and then to call them by name. This would build the chunking operation right into the language. Instead of there being a determinate repertoire of instructions out of which all programs had to be explicitly assembled, the programmer could construct his own modules, each with its own name, each usable anywhere inside the program, just as if it had been a built-in feature of the language. Of course, there is no getting away from the fact that down below, on a machine language level, everything would still be composed of the same old machine language instructions, but that would not be explicitly visible to the highlevel programmer; it would be implicit.

The new languages based on these ideas were called compiler languages. One of the earliest and most elegant was called "Algol", for "Algorithmic Language". Unlike the case with assembly language, there is no straightforward one-to-one correspondence between statements in Algol and machine language instructions. To be sure, there is still a type of mapping from Algol into machine language, but it is far more "scrambled" than that between assembly language and machine language. Roughly speaking, an Algol program is to its machine language translation as a word problem in an elementary algebra text is to the equation it translates into. (Actually, getting from a word problem to an equation is far more complex, but it gives some inkling of the types of "unscrambling" that have to be carried out in translating from a high-level language to a lower-level Ian

guage.) In the mid-1950's, successful programs called compilers were written whose function was to carry out the translation from compiler languages to machine language.

Also, interpreters were invented. Like compilers, interpreters translate from high-level languages into machine language, but instead of translating all the statements first and then executing the machine code, they read one line and' execute it immediately. This has the advantage that a user need not have written a complete program to use an interpreter. He may invent his program line by line, and test it out as he goes along. Thus, an interpreter is to a compiler as a simultaneous interpreter is to a translator of a written speech. One of the most important and fascinating of all computer languages is **LISP** (standing for "List Processing"), which was invented by John McCarthy around the time Algol was invented. Subsequently, **LISP** has enjoyed great popularity with workers in Artificial Intelligence.

There is one interesting difference between the way interpreters work and compilers work. A compiler takes input (a finished Algol program, for instance) and produces output (a long sequence of machine language instructions). At this point, the compiler has done its duty. The output is then given to the computer to run. By contrast, the interpreter is constantly running while the programmer types in one **LISP** statement after another, and each one gets executed then' and there. But this doesn't mean that each statement gets first translated, then executed, for then an interpreter would be nothing but a line-by-line compiler. Instead, in an interpreter, the operations of reading a new line, "understanding" it, and executing it are intertwined: they occur simultaneously.

Here is the idea, expanded a little more. Each time a new line of **LISP** is typed in, the interpreter tries to process it. This means that the interpreter jolts into action, and certain (machine language) instructions inside it get executed. Precisely which ones get executed depends on the **LISP** statement itself, of course. There are many **JUMP** instructions inside the interpreter, so that the new line of **LISP** may cause control to move around in a complex way-forwards, backwards, then forwards again, etc.. Thus, each **LISP** statement gets converted into a "pathway" inside the interpreter, and the act of following that pathway achieves the desired effect.

Sometimes it is helpful to think of the **LISP** statements as mere pieces of data which are fed sequentially to a constantly running machine language program (the **LISP** interpreter). When you think of things this way, you get a different image of the relation between a program written in a higher-level language and the machine which is executing it.

Bootstrapping

Of course a compiler, being itself a program, has to be written in some language. The first compilers were written in assembly language, rather than machine language, thus taking full advantage of the already ac-

omplished first step up from machine language. A summary of these rather tricky concepts is presented in Figure 58.

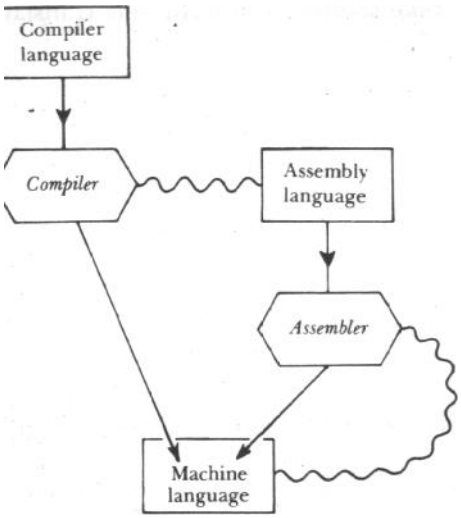


FIGURE 58. Assemblers and compilers are both translators into machine language. This is indicated by the solid lines. Moreover, since they are themselves programs, they are originally written in a language also. The wavy lines indicate that aa compiler can be written in assembly language, and an assembler in machine language.

Now as sophistication increased, people realized that a partially written compiler could be used to compile extensions of itself. In other words, once i certain minimal core of a compiler had been written, then that minimal compiler could translate bigger compilers into machine language-which n turn could translate yet bigger compilers, until the final, full-blown :compiler had been compiled. This process is affectionately known as `bootstrapping"-for obvious reasons (at least if your native language is English it is obvious). It is not so different from the attainment by a child of a critical level of fluency in his native language, from which point on his 'vocabulary and fluency can grow by leaps and bounds, since he can use language to acquire new language.

Levels on Which to Describe Running Programs

Compiler languages typically do not reflect the structure of the machines which will run programs written in them. This is one of their chief advantages over the highly specialized assembly and machine languages. Of course, when a compiler language program is translated into machine language, the resulting program is machine-dependent. Therefore one can describe a program which is being executed in a machine-independent way or a machine-dependent way. It is like referring to a paragraph in a book by its subject matter (publisher-independent), or its page number and position on the page (publisher-dependent).

As long as a program is running correctly, it hardly matters how you describe it or think of its functioning. It is when something goes wrong that

it is important to be able to think on different levels. If, for instance, the machine is instructed to divide by zero at some stage, it will come to a halt and let the user know of this problem, by telling where in the program the questionable event occurred. However, the specification is often given on a lower level than that in which the programmer wrote the program. Here are three parallel descriptions of a program grinding to a halt:

Machine Language Level:

> "Execution of the program stopped in location 1110010101110111"

Assembly Language Level•:

> "Execution of the program stopped when the DIV (divide) instruction was hit"

Compiler Language Level:

> "Execution of the program stopped during evaluation of the algebraic expression `(A + B)/Z'

One of the greatest problems for systems programmers (the people who write compilers, interpreters, assemblers, and other programs to be used by many people) is to figure out how to write error-detecting routines in such a way that the messages which they feed to the user whose program has a "bug" provide high-level, rather than low-level, descriptions of the problem. It is an interesting reversal that when something goes wrong in a genetic "program" (e.g., a mutation), the "bug" is manifest only to people on a high level-namely on the phenotype level, not the genotype level. Actually, modern biology uses mutations as one of its principal windows onto genetic processes, because of their multilevel traceability.

Microprogramming and Operating Systems

In modern computer systems, there are several other levels of the hierarchy. For instance, some systems-often the so-called "microcomputers" come with machine language instructions which are even more rudimentary than the instruction to add a number in memory to a number in a register. It is up to the user to decide what kinds of ordinary machine-level instructions he would like to be able to program in; he "microprograms" these instructions in terms of the "micro-instructions" which are available. Then the "higher-level machine language" instructions which he has designed may be burned into the circuitry and become hard-wired, although they need not be. Thus microprogramming allows the user to step a little below the conventional machine language level. One of the consequences is that a computer of one manufacturer can be hard-wired (via microprogramming) so as to have the same machine language instruction set as a computer of the same, or even another, manufacturer. The microprogrammed computer is said to be "emulating" the other computer. Then there is the level of the *operating system*, which fits between the

machine language program and whatever higher level the user is programming in. The operating system is itself a program which has the functions of shielding the bare machine from access by users (thus protecting the system), and also of insulating the programmer from the many extremely intricate and messy problems of reading the program, calling a translator, running the translated program, directing the output to the proper channels at the proper time, and passing control to the next user. If there are several users "talking" to the same CPU at once, then the operating system is the program that shifts attention from one to the other in some orderly fashion. The complexities of operating systems are formidable indeed, and I shall only hint at them by the following analogy.

Consider the first telephone system. Alexander Graham Bell could phone his assistant in the next room: electronic transmission of a voice! Now that is like a bare computer minus operating system: electronic computation! Consider now a modern telephone system. You have a choice of other telephones to connect to. Not only that, but many different calls can be handled simultaneously. You can add a prefix and dial into different areas. You can call direct, through the operator, collect, by credit card, person-to-person, on a conference call. You can have a call rerouted or traced. You can get a busy signal. You can get a siren-like signal that says that the number you dialed isn't "well-formed", or that you have taken too in long in dialing. You can install a local switchboard so that a group of phones are all locally connected--etc., etc. The list is amazing, when you think of how much flexibility there is, particularly in comparison to the erstwhile miracle of a "bare" telephone. Now sophisticated operating systems carry out similar traffic-handling and level-switching operations with respect to users and their programs. It is virtually certain that there are somewhat parallel things which take place in the brain: handling of many stimuli at the same time; decisions of what should have priority over what and for how long; instantaneous "interrupts" caused by emergencies or other unexpected occurrences; and so on.

Cushioning the User and Protecting the System

The many levels in a complex computer system have the combined effect of "cushioning" the user, preventing him from having to think about the many lower-level goings-on which are most likely totally irrelevant to him anyway. A passenger in an airplane does not usually want to be aware of the levels of fuel in the tanks, or the wind speeds, or how many chicken dinners are to be served, or the status of the rest of the air traffic around the destination-this is all left to employees on different levels of the airlines hierarchy, and the passenger simply gets from one place to another. Here again, it is when something goes wrong-such as his baggage not arriving that the passenger is made aware of the confusing system of levels underneath him.

Are Computers Super-Flexible or Super-Rigid?

One of the major goals of the drive to higher levels has always been to make as natural as possible the task of communicating to the computer what you want it to do. Certainly, the high-level constructs in compiler languages are closer to the concepts which humans naturally think in, than are lower-level constructs such as those in machine language. But in this drive towards ease of communication, one aspect of "naturalness" has been quite neglected. That is the fact that interhuman communication is far less rigidly constrained than human-machine communication. For instance, we often produce meaningless sentence fragments as we search for the best way to express something, we cough in the middle of sentences, we interrupt each other, we use ambiguous descriptions and "improper" syntax, we coin phrases and distort meanings-but our message still gets through, mostly. With programming languages, it has generally been the rule that there is a very strict syntax which has to be obeyed one hundred per cent of the time; there are no ambiguous words or constructions. Interestingly, the printed equivalent of coughing (i.e., a nonessential or irrelevant comment) is allowed, but only provided it is signaled in advance by a key word (e.g., **COMMENT**), and then terminated by another key word (e.g., a semicolon). This small gesture towards flexibility has its own little pitfall, ironically: if a semicolon (or whatever key word is used for terminating a comment) is used inside a comment, the translating program will interpret that semicolon as signaling the end of the comment, and havoc will ensue.

If a procedure named **INSIGHT** has been defined and then called seventeen times in the program, and the eighteenth time it is misspelled as **INSIHGT**, woe to the programmer. The compiler will balk and print a rigidly unsympathetic error message, saying that it has never heard of **INSIHGT**. Often, when such an error is detected by a compiler, the compiler tries to continue, but because of its lack of insihgt, it has not understood what the programmer meant. In fact, it may very well suppose that something entirely different was meant, and proceed under that erroneous assumption. Then a long series of error messages will pepper the rest of the program, because the compiler-not the programmer-got confused. Imagine the chaos that would result if a simultaneous English-Russian interpreter, upon hearing one phrase of French in the English, began trying to interpret all the remaining English as French. Compilers often get lost in such pathetic ways. C'est la vie.

Perhaps this sounds condemnatory of computers, but it is not meant to be. In some sense, things had to be that way. When you stop to think what most people use computers for, you realize that it is to carry out very definite and precise tasks, which are too complex for people to do. If the computer is to be reliable, then it is necessary that it should understand, without the slightest chance of ambiguity, what it is supposed to do. It is also necessary that it should do neither more nor less than it is explicitly instructed to do. If there is, in the cushion underneath the programmer, a program whose purpose is to "guess" what the programmer wants or

means, then it is quite conceivable that the programmer could try to communicate his task and be totally misunderstood. So it is important that the high-level program, while comfortable for the human, still should be unambiguous and precise.

Second-Guessing the Programmer

Now it is possible to devise a programming language-and a program which translates it into the lower levels-which allows some sorts of imprecision. One way of putting it would be to say that a translator for such a programming language tries to make sense of things which are done "outside of the rules of the language". But if a language allows certain "transgressions", then transgressions of that type are no longer true transgressions, because they have been included inside the rules' If a programmer is aware that he may make certain types of misspelling, then he may use this feature of the language deliberately, knowing that he is actually operating within the rigid rules of the language, despite appearances. In other words, if the user is aware of all the flexibilities programmed into the translator for his convenience, then he knows the bounds which he cannot overstep, and therefore, to him, the translator still appears rigid and inflexible, although it may allow him much more freedom than early versions of the language, which did not incorporate "automatic compensation for human error".

With "rubbery" languages of that type, there would seem to be two alternatives: (1) the user is aware of the built-in flexibilities of the language and its translator; (2) the user is unaware of them. In the first case, the language is still usable for communicating programs precisely, because the programmer can predict how the computer will interpret the programs he writes in the language. In the second case, the "cushion" has hidden features which may do things that are unpredictable (from the vantage point of a user who doesn't know the inner workings of the translator). This may result in gross misinterpretations of programs, so such a language is unsuitable for purposes where computers are used mainly for their speed and reliability.

Now there is actually a third alternative: (3) the user is aware of the built-in flexibilities of the language and its translator, but there are so many of them and they interact with each other in such a complex way that he cannot tell how his programs will be interpreted. This may well apply to the person who wrote the translating program; he certainly knows its insides as well as anyone could-but he still may not be able to anticipate how it will react to a given type of unusual construction.

One of the major areas of research in Artificial Intelligence today is called automatic programming, which is concerned with the development of yet higher-level languages-languages whose translators are sophisticated. in that they can do at least some of the following impressive things: generalize from examples, correct some misprints or grammatical errors,

try to make sense of ambiguous descriptions, try to second-guess the user by having a primitive user model, ask questions when things are unclear, use English itself, etc. The hope is that one can walk the tightrope between reliability and flexibility.

AI Advances Are Language Advances

It is striking how tight the connection is between progress in computer science (particularly Artificial Intelligence) and the development of new languages. A clear trend has emerged in the last decade: the trend to consolidate new types of discoveries in new languages. One key for the understanding and creation of intelligence lies in the constant development and refinement of the languages in terms of which processes for symbol manipulation are describable. Today, there are probably three or four dozen experimental languages which have been developed exclusively for Artificial Intelligence research. It is important to realize that any program which can be written in one of these languages is in principle programmable in lower-level languages, but it would require a supreme effort for a human; and the resulting program would be so long that it would exceed the grasp of humans. It is not that each higher level extends the potential of the computer; the full potential of the computer already exists in its machine language instruction set. It is that the new concepts in a high-level language suggest directions and perspectives by their very nature.

The "space" of all possible programs is so huge that no one can have a sense of what is possible. Each higher-level language is naturally suited for exploring certain regions of "program space"; thus the programmer, by using that language, is channeled into those areas of program space. He is not forced by the language into writing programs of any particular type, but the language makes it easy for him to do certain kinds of things. Proximity to a concept, and a gentle shove, are often all that is needed for a major discovery-and that is the reason for the drive towards languages of ever higher levels.

Programming in different 'languages is like composing pieces in different keys, particularly if you work at the keyboard. If you have learned or written pieces in many keys, each key will have its own special emotional aura. Also, certain kinds of figurations "lie in the hand" in one key but are awkward in another. So you are channeled by your choice of key. In some ways, even enharmonic keys, such as C-sharp and D-flat, are quite distinct in feeling. This shows how a notational system can play a significant role in shaping the final product.

A "stratified" picture of **Al** is shown in Figure 59, -with machine components such as transistors on the bottom, and "intelligent programs" on the top. The picture is taken from the book *Artificial Intelligence* by Patrick Henry Winston, and it represents a vision of **Al** shared by nearly all Al workers. Although I agree with the idea that **Al** must be stratified in some such way, I do not think that, with so few layers, intelligent programs

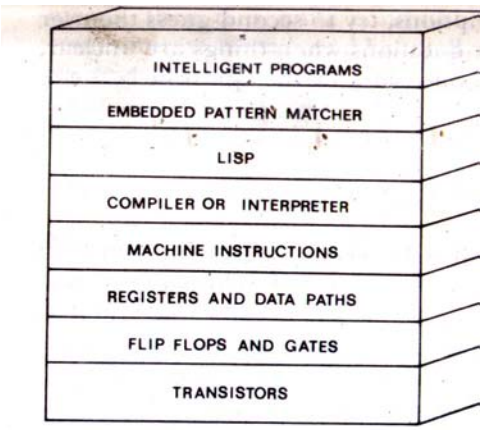


FIGURE 59. To create intelligent programs, one needs to build up a series of levels of hardware and software, so that one is spared the agonT of seeing everything only on the lowest level. Descriptions of a single process on different levels will sound verb different from each other, only the top one being sufficiently chunked that it is comprehensible to us. [Adapted from P. H. Winston, Artificial Intelligence (Reading, Mass.: Addison-ifele", 1977)]

can he reached. Between the machine language level and the level where rue intelligence will be reached, I am convinced there will lie perhaps mother dozen (or even several dozen!) layers, each new layer building on and extending the flexibilities of the layer below. What they will be like we can hardly dream of now ...

The Paranoid and the Operating System

The similarity of all levels in a computer system can lead to some strange level-mixing experiences. I once watched a couple of friends-both computer novices-playing with the program "PARRY" on a terminal. PARRY s a. rather infamous program which simulates a paranoid in an extremely rudimentary way, by spitting out canned phrases in English chosen from a vide repertoire; its plausibility is due to its ability to tell which of its stock phrases might sound reasonable in response to English sentences typed to t by a human.

At one point, the response time got very sluggish-PARRY was taking very long to reply-and I explained to my friends that this was probably because of the heavy load on the time-sharing system. I told them they could find out how many users were logged on, by typing a special "control" character which would go directly to the operating system, and would)e unseen by PARRY. One of my friends pushed the control character. In a lash, some internal data about the operating system's status overwrote some of PARRY's words on the screen. PARRY knew nothing of this: it is a program with "knowledge" only of horse racing and bookies-not operating systems and terminals and special control characters. But to my friends, both PARRY and the operating system were just "the computer"-a mysterious, remote, amorphous entity that responded to them when they typed. And so it made perfect sense when one of them blithely typed, in 3nglish, "Why are you overtyping what's on the screen?" The idea that PARRY could know' nothing about the operating system it was running

under was not clear to my friends. The idea that "you" know all about "yourself" is so familiar from interaction with people that it was natural to extend it to the computer-after all, it was intelligent enough that it could "talk" to them in English! Their question was not unlike asking a person, "Why are you making so few red blood cells today?" People do not know about that level-the "operating system level"-of their bodies.

The main cause of this level-confusion was that communication with all levels of the computer system was taking place on a single screen, on a single terminal. Although my friends' naiveté might seem rather extreme, even experienced computer people often make similar errors when several levels of a complex system are all present at once on the same screen. They forget "who" they are talking to, and type something which makes no sense at that level, although it would have made perfect sense on another level. It might seem desirable, therefore, to have the system itself sort out the levels-to interpret commands according to what "makes sense". Unfortunately, such interpretation would require the system to have a lot of common sense, as well as perfect knowledge of the programmer's overall intent-both of which would require more artificial intelligence than exists at the present time.

The Border between Software and Hardware

One can also be confused by the flexibility of some levels and the rigidity of others. For instance, on some computers there are marvelous text-editing systems which allow pieces of text to be "poured" from one format into another, practically as liquids can be poured from one vessel into another. A thin page can turn into a wide page, or vice versa. With such power, you might expect that it would be equally trivial to change from one font to another-say from roman to italics. Yet there may be only a single font available on the screen, so that such changes are impossible. Or it may be feasible on the screen but not printable by the printer-or the other way around. After dealing with computers for a long time, one gets spoiled, and thinks that everything should be programmable: no printer should be so rigid as to have only one character set, or even a finite repertoire of them-typefaces should be user-specifiable! But once that degree of flexibility has been attained, then one may be annoyed that the printer cannot print in different colors of ink, or that it cannot accept paper of all shapes and sizes, or that it does not fix itself when it breaks ...

The trouble is that somewhere, all this flexibility has to "bottom out", to use the phrase from Chapter V. There must be a hardware level which underlies it all, and which is inflexible. It may lie deeply hidden, and there may be so much flexibility on levels above it that few users feel the hardware limitations-but it is inevitably there.

What is this proverbial distinction between software and hardware? It is the distinction between programs and machines-between long complicated sequences of instructions, and the physical machines which carry

them out. I like to think of software as "anything which you could send over he telephone lines", and hardware as "anything else". A piano is hardware, gut printed music is software. A telephone set is hardware, but a telephone lumber is software. `The distinction is a useful one, but not always so clear-cut.

We humans also have "software" and "hardware" aspects, and the difference is second nature to us. We are used to the rigidity of our physiology: the fact that we cannot, at will, cure ourselves of diseases, or ;row hair of any color-to mention just a couple of simple examples. We an, however, "reprogram" our minds so that we operate in new conceptual frameworks. The amazing flexibility of our minds seems nearly irreconcilable with the notion that our brains must be made out of fixed-rule hardware, which cannot be reprogrammed. We cannot make our neurons ire faster or slower, we cannot rewire our brains, we cannot redesign the interior of a neuron, we cannot make anti choices about the hardware-and 'et, we can control how we think.

But there are clearly aspects of thought which are beyond our control. We cannot make ourselves smarter by an act of will; we cannot learn a new language as fast as we want; we cannot make ourselves think faster than we lo; we cannot make ourselves think about several things at once; and so on. This is a kind of primordial self-knowledge which is so obvious that it is lard to see it at all; it is like being conscious that the air is there. We never really bother to think about what might cause these "defects" of our minds: lamely, the organization of our brains. To suggest ways of reconciling the software of mind with the hardware of brain is a main goal of this book.

Intermediate Levels and the Weather

We have seen that in computer systems, there are a number of rather sharply defined strata, in terms of any one of which the operation of a running program can be described. Thus there is not merely a single low bevel and a single high level-there are all degrees of lowness and highness. s the existence of intermediate levels a general feature of systems which lave low and high levels? Consider, for example, the system whose 'hardware" is the earth's atmosphere (not very hard, but no matter), and whose "software" is the weather. Keeping track of the motions of all of the molecules simultaneously would be a very low-level way of "understanding" he weather, rather like looking at a huge, complicated program on the machine language level. Obviously it is way beyond human comprehension. 3ut we still have our own peculiarly human ways of looking at, and describing, weather phenomena. Our chunked view of the weather is based >n very high-level phenomena, such as: rain, fog, snow, hurricanes, cold fronts, seasons, pressures, trade winds, the jet stream, cumulo-nimbus clouds, thunderstorms, inversion layers, and so on. All of these phenomena involve astronomical numbers of molecules, somehow behaving in concert o that large-scale trends emerge. This is a little like looking at the weather n a compiler language.

Is there something analogous to looking at the weather in an intermediate-level language, such as assembly language? For instance, are there very small local "mini-storms", something like the small whirlwinds which one occasionally sees, whipping up some dust in a swirling column a few feet wide, at most? Is a local gust of wind an intermediate-level chunk which plays a role in creating higher-level weather phenomena? Or is there just no practical way of combining knowledge of such kinds of phenomena to create a more comprehensive explanation of the weather?

Two other questions come to my mind. The first is: "Could it be that the weather phenomena which we perceive on our scale-a tornado, a drought-are just intermediate-level phenomena: parts of vaster, slower phenomena?" If so, then true high-level weather phenomena would be global, and their time scale would be geological. The Ice Age would be a high-level weather event. The second question is: "Are there intermediate level weather phenomena which have so far escaped human perception, but which, if perceived, could give greater insight into why the weather is as it is?"

From Tornados to Quarks

This last suggestion may sound fanciful, but it is not all that far-fetched. We need only look to the hardest of the hard sciences-physics-to find peculiar examples of systems which are explained in terms of interacting "parts" which are themselves invisible. In physics, as in any other discipline, a system is a group of interacting parts. In most systems that we know, the parts retain their identities during the interaction, so that we still see the parts inside the system. For example, when a team of football players assembles, the individual players retain their separateness-they do not melt into some composite entity, in which their individuality is lost. Still-and this is important-some processes are going on in their brains which are evoked by the team-context, and which would not go on otherwise, so that in a minor way, the players change identity when they become part of the larger system, the team. This kind of system is called a nearly decomposable system (the term comes from H. A. Simon's article "The Architecture of Complexity"; see the Bibliography). Such a system consists of weakly interacting modules, each of which maintains its own private identity throughout the interaction but by becoming slightly different from how it is when outside of the system,, contributes to the cohesive behavior of the whole system. The systems studied in physics are usually of this type. For instance, an atom is seen as made of 'a nucleus whose positive charge captures a number of electrons in "orbits", or bound states. The bound electrons are very much like free electrons, despite their being internal to a composite object.

Some systems studied in physics offer a contrast to the relatively straightforward atom. Such systems involve extremely strong interactions, as a result of which the parts are swallowed up into the larger system, and lose some or all of their individuality. An example of this is the nucleus of an atom, which is usually described as being "a collection of protons and

neutrons". But the forces which pull the component particles together are strong that the component particles do not survive to anything like their “free" form (the form they have when outside a nucleus). And in fact a nucleus acts in many ways as a single particle, rather than as a collection of interacting particles. When a nucleus is split. protons and neutrons are ten released. but also other particles. such as pi-mesons and gamma rays, are commonly produced. Are all those different particles physically present side a nucleus before it is split, or are then just "sparks" which fly off ten the nucleus is split- It is perhaps not meaningful to try to give an answer to such a question. On the level of particle physics, the difference between storing the potential to make "sparks" and storing actual sub particles is not so clear.

A nucleus is thus one systems whose "parts!, even though they are not visible while on the inside, can be pulled out and made risible. However, ere are more pathological cases. such as the proton and neutron seen as stems themselves. Each of them has been hypothesized to be constituted from a trio of "quarks"- hypothetical particles which can be combined in twos or threes to make many known fundamental particles. However, the interaction between quarks is so strong that not only can they not he seen [side the proton and neutron, but they cannot even be pulled out at all'. bus, although quarks help to give a theoretical understanding of certain properties of protons and neutrons, their own existence may perhaps ever be independently established. Here see have the antithesis of a nearly decomposable system"-it is a system which, if anything, is "nearly indecomposable", Yet what is curious is that a quark-based theory of rotors and neutrons (and other particles) has considerable explanatory power. in that many experimental results concerning the particles which narks supposedly compose can be accounted for quite well, quantitatively. by using the "quark model".

Superconductivity: A "Paradox" of Renormalization

In Chapter V we discussed how renormalized particles emerge from their bare cores, by recursively compounded interactions with virtual particles. A renormalized particle can be seen either as this complex mathematical construct, or as the single lump which it is, physically. One of the strangest rid most dramatic consequences of this way of describing particles is the explanation it provides for the famous phenomenon of *superconductivity* resistance-free flow of electrons in certain solids, at extremely low temperatures.

It turns out that electrons in solids are renormalized by their interactions with strange quanta of vibration called *phonons* (themselves renormalized as well!). These renormalized electrons are called *polarons*. Calculation shows that at very low temperatures, two oppositely spinning polarons sill begin to attract each other, and can actually become bound together in i certain way. Under the proper conditions. all the current-carrying polar

ons will get paired up, forming Cooper pains. Ironically, this pairing comes about precisely because electrons-the hare cores of the paired polarons--repel each other electrically. In contrast to the electrons, each Cooper pair feels neither attracted to nor repelled by an other Cooper pair, and consequently it can slip freely through a metal as if the metal were a vacuum. If you convert the mathematical description of such a metal from one whose primitive units are polarons into one whose primitive units are Cooper pairs. you get a considerable- simplified set of equations. This mathematical simplicity is the physicist's way of knowing that "chunking" into Cooper pairs is the natural way to look at superconductivity.

Here we have several levels of particle: the Cooper pair itself: the two oppositely-spinning polarons which compose it: the electrons and phonons which make up the polarons: and then, within the electrons, the virtual photons and positrons, etc. etc. We can look at each level and perceive phenomena there, which are explained by an understanding of the levels below.

"Sealing-off"

Similarly, and fortunately. one does not have to know all about quarks to understand many things about the particles which they may compose. Thus, a nuclear physicist can proceed with theories of nuclei that are based on protons and neutrons, and ignore quark theories and their rivals. The nuclear physicist has a chunked picture of protons and neutrons-a description derived from lower-level theories buf which does not require understanding the lower-level theories. Likewise, an atomic physicist has a chunked picture of an atomic nucleus derived from nuclear theory. Then a chemist has a chunked picture of the electrons and their orbits, and builds theories of small molecules, theories which can be taken over in a chunked way by the molecular biologist, who has an intuition for how small molecules hang together, but whose technical expertise is in the field of extremely large molecules and how they interact. Then the cell biologist has a chunked picture of the units which the molecular biologist pores over, and tries to use them to account f'or the ways that cells interact. The point is clear. Each level is, in some sense, "sealed off" from the levels below it. This is another of Simon's vivid terms, recalling the way in which a submarine is built in compartments, so that if one part is damaged, and water begins pouring in, the trouble can be prevented from spreading, by closing the doors, thereby sealing off the damaged compartment from neighboring compartments.

Although there is always some "leakage" between the hierarchical levels of science, so that a chemist cannot afford to ignore lower-level physics totally, or a biologist to ignore chemistry totally, there is almost no leakage from one level to a distant level. That is why people earl, have intuitive understandings of other people without necessarily understanding the quark model, the structure of nuclei, the nature of electron orbits,

the chemical bond, the structure of proteins, the organelles in a cell, the methods of intercellular communication, the physiology 'of the various organs of the human body, or the complex interactions among organs. All at a person needs is a chunked model of how the highest level acts; and as all know, such models are very realistic and successful.

The Trade-off between Chunking and Determinism

There is, however, perhaps one significant negative feature of a chunked model: it usually does not have exact predictive power. That is, we save ourselves from the impossible task of seeing people as collections of quarks (or whatever is at the lowest level) by using chunked models: but of course such models only give us probabilistic estimates of how other people feel, wil1 react to what we say or do, and so on. In short, in using chunked high-level models, we sacrifice determinism for simplicity. Despite not being sure how people will react to a joke, we tell it with the expectation at they will do something such as laugh, or not laugh-rather than, say, climb the nearest flagpole. (Zen masters might well do the latter!) A chunked model defines a "space" within which behavior is expected to fall, and specifies probabilities of its falling in different parts of that space.

"Computers Can Only Do What You Tell Them to Do"

Now these ideas can be applied as well to computer programs as to compose physical systems. There is an old saw which says, "Computers can only what you tell them to do." This is right in one sense, but it misses the hint: you don't know in advance the consequences of what you tell a computer to do; therefore its behavior can be as baffling and surprising id unpredictable to you as that of a person. You generally know in advance the space in which the output will fall, but you don't know details of here it will fall. For instance, you might write a program to calculate the first million digits of 7r. Your program will spew forth digits of 7r much faster than you can-but there is no paradox in the fact that the computer outracing its programmer. You know in advance the space in which the output will lie-namely the space of digits between 0 and 9-which is to say,)u have a chunked model of the program's behavior; but if you'd known ie rest, you wouldn't have written the program.

There is another sense in which this old saw is rusty. This involves the ct that as you program in ever higher-level languages, you know less and ss precisely what you've told the computer to do! Layers and layers of translation may separate the "front end" of a complex program from the actual machine language instructions. At the level you think and program, your statements may resemble declaratives and suggestions more than they resemble imperatives or commands. And all the internal rumbling provoked by the input of a high-level statement is invisible to you, generally, just as when you eat a sandwich, you are spared conscious awareness of the digestive processes it triggers

In any case, this notion that "computers can only do what they are told to do," first propounded by Lady Lovelace in her famous memoir, is so prevalent and so connected with the notion that "computers cannot think" that we shall return to it in later Chapters when our level of sophistication is greater.

Two Types of System

There is an important division between two types of system built up from many parts. There are those systems in which the behavior of some parts tends to cancel out the behavior of other parts, with the result that it does not matter too much what happens on the low level, because most anything will yield similar high-level behavior. An example of this kind of system is a container of gas, where all the molecules bump and bang against each other in very complex microscopic ways; but the total outcome, from a macroscopic point of view, is a very calm, stable system with a certain temperature, pressure, and volume. Then there are systems where the effect of a single low-level event may get magnified into an enormous high-level consequence. Such a system is a pinball machine, where the exact angle with which a ball strikes each post is crucial in determining the rest of its descending pathway.

A computer is an elaborate combination of these two types of system. It contains subunits such as wires, which behave in a highly predictable fashion: they conduct electricity according to Ohm's law, a very precise, chunked law which resembles the laws governing gases in containers, since it depends on statistical effects in which billions of random effects cancel each other out, yielding a predictable overall behavior. A computer also contains macroscopic subunits, such as a printer, whose behavior is completely determined by delicate patterns of currents. What the printer prints is not by any means created by a myriad canceling microscopic effects. In fact, in the case of most computer programs, the value of every single bit in the program plays a critical role in the output that gets printed. If any bit were changed, the output would also change drastically.

Systems which are made up of "reliable" subsystems only-that is, subsystems whose behavior can be reliably predicted from chunked descriptions-play inestimably important roles in our daily lives, because they are pillars of stability. We can rely on walls not to fall down, on sidewalks to go where they went yesterday, on the sun to shine, on clocks to tell the time correctly, and so on. Chunked models of such systems are virtually entirely deterministic. Of course, the other kind of system which plays a very large role in our lives is a system that has variable behavior which depends on some internal microscopic parameters-often a very large number of them, moreover-which we cannot directly observe. Our chunked model of such a system is necessarily in terms of the "space" of operation, and involves probabilistic estimates of landing in different regions of that space.

A container of gas, which, as I already pointed out, is a reliable system

Because of many canceling effects, obeys precise, deterministic laws of physics. Such laws are chunked laws, in that they deal with the gas as a whole, nd ignore its constituents. Furthermore, the microscopic and macroscopic descriptions of a gas use entirely different terms. The former requires the pacification of the position and velocity of every single component molecule; the latter requires only the specification of three new quantities: temperature, pressure, and volume, the first two of which do not even have microscopic counterparts. The simple mathematical relationship which elates these three parameters- pV = cT, where c is a constant-is a law which depends on, yet is independent of, the lower-level phenomena. Less paradoxically, this law can be derived from the laws governing the molecular level; in that sense it depends on the lower level. On the other hand, it is law which allows you to ignore the lower level completely, if you wish: in hat sense it is independent of the lower level.

It is important to realize that the high-level law cannot be stated in the vocabulary of the low-level description. "Pressure" and "temperature" are new terms which experience with the low level alone cannot convey. We humans perceive temperature and pressure directly; that is how we are guilt, so that it is not amazing that we should have found this law. But creatures which knew gases only as theoretical mathematical constructs would have to have an ability to synthesize new concepts, if they were to discover this law.

Epiphenomena

In drawing this Chapter to a close, I would like to relate a story about a complex system. I was talking one day with two systems programmers for he computer I was using. They mentioned that the operating system seemed to be able to handle up to about thirty-five users with great comfort, but at about thirty-five users or so, the response time all of a sudden hot up, getting so slow that you might as well log off and go home and wait until later. Jokingly I said, "Well, that's simple to fix just find the place in he operating system where the number `35' is stored, and change it to 60'!" Everyone laughed. The point is, of course, that there is no such place. where, then, does the critical number-35 users-come from? The answer is: *It is a visible consequence of the overall system organization*-an "*epiphenometon*,,.

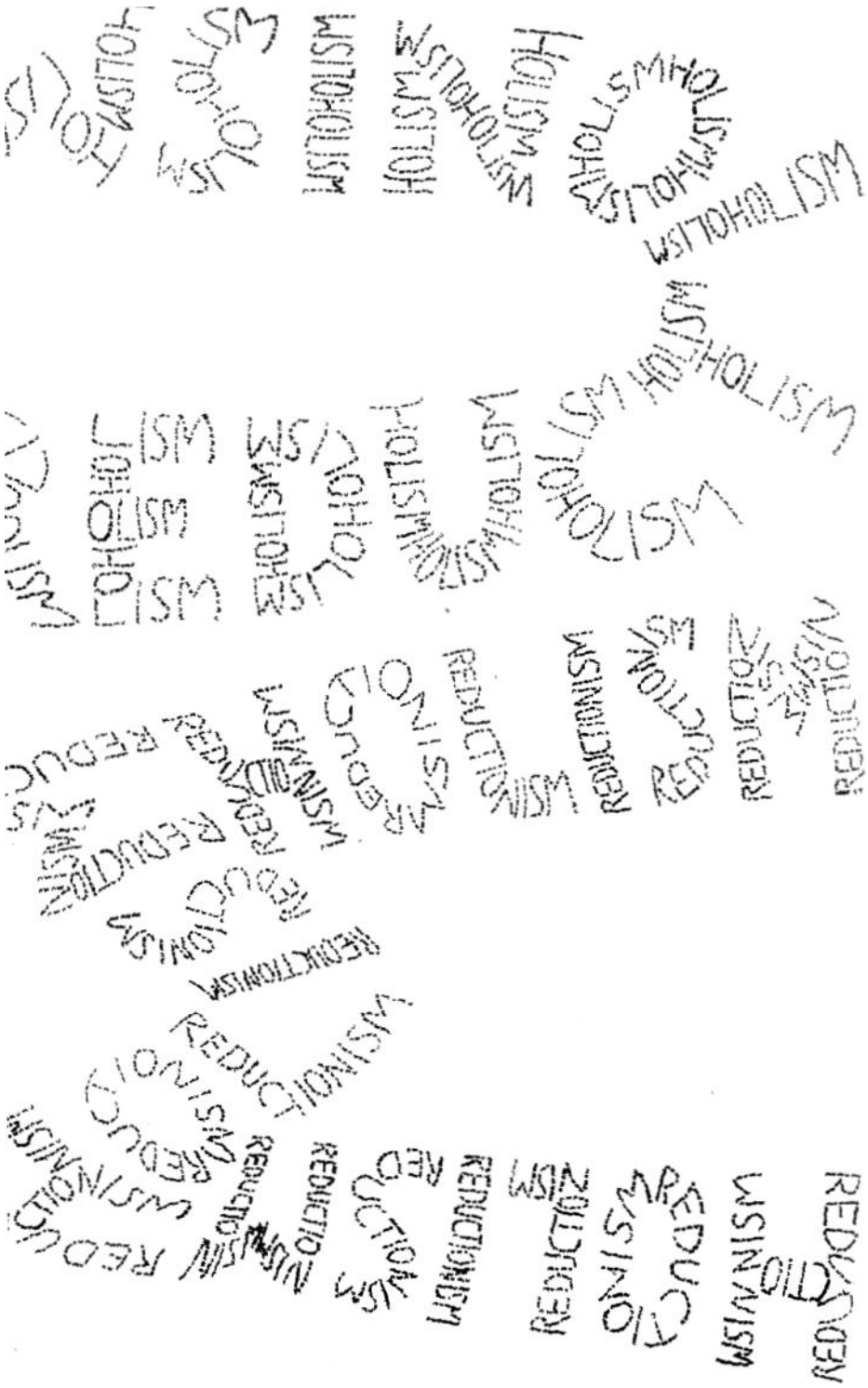
Similarly, you might ask about a sprinter, "Where is the `9.3' stored, hat makes him be able to run 100 yards in 9.3 seconds?" Obviously, it is not stored anywhere. His time is a result of how he is built, what his reaction time is, a million factors all interacting when he runs. The time is quite 'reproducible, but it is not stored in his body anywhere. It is spread around among all the cells of his body and only manifests itself in the act of the print itself.

Epiphenomena abound. In the game of "Go", there is the feature that “two eyes live”. It is not built into the rules, but it is a consequence of the

rules. In the human brain, there is gullibility. How gullible are you? Is your gullibility located in some "gullibility center" in your brain? Could a neurosurgeon reach in and perform some delicate operation to lower your gullibility, otherwise leaving you alone? If you believe this, you are pretty gullible, and should perhaps consider such an operation.

Mind *vs*. Brain

In coming Chapters, where we discuss the brain, we shall examine whether the brain's top level-the mind-can be understood without understanding the lower levels on which it both depends and does not depend. Are there laws of thinking which are "sealed off" from the lower laws that govern the microscopic activity in the cells of the brain? Can mind be "skimmed" off of brain and transplanted into other systems? Or is it impossible to unravel thinking processes into neat and modular subsystems? Is the brain more like an atom, a renormalized electron, a nucleus, a neutron, or a quark? Is consciousness an epiphenomenon? To understand the mind, must one go all the way down to the level of nerve cells?


REDUCTIONISM
REDUCTIONISM
REDUCTIONISM

... Ant Fugue

. . . . then, one by one, the four voices of the fugue chime in.)

Achilles: I know the rest of you won't believe this, but the answer to the question is staring us all in the face, hidden in the picture. It is simply one word-but what an important one: "MU"!

CCrab: I know the rest of you won't believe this, but the answer to the question is staring us all in the face, hidden in the picture. It is simply one word-but what an important one: "HOLISM"!

Achilles: Now hold on a minute. You must be seeing things. It's plain as day that the message of this picture is "MU", not "HOLISM"!

Crab: I beg your pardon, but my eyesight is extremely good. Please look again, and then tell me if the the picture doesn't say what I said it says!

Anteater: I know the rest of you won't believe this, but the answer to the question is staring us all in the face, hidden in the picture. It is simply one word-but what an important one: "REDUCTIONISM"!

Crab: Now hold on a minute. You must be seeing things. It's plain as day that the message of this picture is "HOLISM", not "REDUCTIONISM"!

Achilles: Another deluded one! Not "HOLISM", not "REDUCTIONISM", but "MU" is the message of this picture, and that much is certain.

Anteater: I beg your pardon, but my eyesight is extremely clear. Please look again, and then see if the picture doesn't say what I said it says.

Achilles: Don't you see that the picture is composed of two pieces, and that each of them is a single letter?

Crab: You are right about the two pieces, but you are wrong in your identification of what they are. The piece on the left is entirely composed of three copies of one word: "HOLISM"; and the piece on the right is composed of many copies, in smaller letters, of the same word. Why the letters are of different sizes in the two parts, I don't know, but I know what I see, and what I see is "HOLISM", plain as day. How you see anything else is beyond me.

Anteater: You are right about the two pieces, but you are wrong in your identification of what they are. The piece on the left is entirely composed of many copies of one word: "REDUCTIONISM"; and the piece on the right is composed of one single copy, in larger letters, of the same word. Why the letters are of different sizes in the two parts, I don't know, but I know what I see, and what I see is "REDUCTIONISM", plain as day. How you see anything else is beyond me.

Achilles: I know what is going on here. Each of you has seen letters which compose, or are composed of, other letters. In the left-hand piece,

there are indeed three "HOLISM"'s, but each one of them is composed out of smaller copies of the word "REDUCTIONISM". And in complementary fashion, in the right-hand piece, there is indeed one "REDUCTIONISM", but it is composed out of smaller copies of the word "HOLISM". Now this is all fine and good, but in your silly squabble, the two of you have actually missed the forest for the trees. You see, what good is it to argue about whether "HOLISM" or "REDUCTIONISM" is right, when the proper way to understand the matter is to transcend the question, by answering "Mu",

Crab: I now see the picture as you have described it, Achilles, but I have no idea of what you mean by the strange expression "transcending the question".

Anteater: I now see the picture as you have described it, Achilles, but I have no idea of what you mean by the strange expression "Mu". .illes: I will be glad to indulge both of you, if you will first oblige me, by telling me the meaning of these strange expressions, "HOLISM" and "REDUCTIONISM".

Crab: HOLISM is the most natural thing in the world to grasp. It's simply the belief that "the whole is greater than the sum of its parts". No one in his right mind could reject holism.

Anteater: REDUCTIONISM is the most natural thing in the world to grasp. It's simply the belief that "a whole can be understood completely if you understand its parts, and the nature of their 'sum'". No one in her left brain could reject reductionism.

Crab: I reject reductionism. I challenge you to tell me, for instance, how to understand a brain reductionistically. Any reductionistic explanation of a brain will inevitably fall far short of explaining where the consciousness experienced by a brain arises from.

Anteater: I reject holism. I challenge you to tell me, for instance, how a holistic description of an ant colony sheds any more light on it than is shed by a description of the ants inside it, and their roles, and their interrelationships. Any holistic explanation of an ant colony will inevitably fall far short of explaining where the consciousness experienced by an ant colony arises from.

Antilles: Oh, no! The last thing which I wanted to do was to provoke another argument. Anyway, now that I understand the controversy, I believe that my explanation of "Mu" will help greatly. You see, "Mu" is an ancient Zen answer which, when given to a question, UNASKS the question. Here, the question seems to be, "Should the world be understood via holism, or via reductionism?" And the answer of "Mu" here rejects the premises of the question, which are that one or the other must be chosen. By unasking the question, it reveals a wider truth: that there is a larger context into which both holistic and reductionistic explanations fit.

Anteater: Absurd! Your "Mu" is as silly as a cow's moo. I'll have none of this Zen washy-wishiness.

Crab: Ridiculous! Your "ML" is as silly as a kitten's mew. I'll have none of this Zen washy-wishiness.

Achilles: Oh, dear! We're getting nowhere fast. Why have you stayed so strangely silent, Mr. Tortoise? It makes me very uneasy. Surely you must somehow be capable of helping straighten out this mess?

Tortoise: I know the rest of you won't believe this, but the answer to the question is staring us all in the face, hidden in the picture. It is simply one word-but what an important one: "Mu"!

Gust as he says this, the fourth voice in the fugue being played enters, exactly one octave below the first entry.)

Achilles: Oh, Mr. T, for once you have let me down. I was sure that you, who always see the most deeply into things, would be able to resolve this dilemma-but apparently, you have seen no further than I myself saw. Oh, well, I guess I should feel pleased to have seen as far as Mr. Tortoise, for once.

Tortoise: I beg your pardon, but my eyesight is extremely fine. Please look again, and then tell me if the picture doesn't say what I said it says.

Achilles: But of course it does! You have merely repeated my own original observation.

Tortoise: Perhaps "Mu" exists in this picture on a deeper level than you imagine, Achilles-an octave lower (figuratively speaking). But for now I doubt that we can settle the dispute in the abstract. I would like to see both the holistic and reductionistic points of view laid out more explicitly; then there may be more of a basis for a decision. I would very much like to hear a reductionistic description of an ant colony, for instance.

Crab: Perhaps Dr. Anteater will tell you something of his experiences in that regard. After all, he is by profession something of an expert on that subject.

Tortoise: I am sure that we have much to learn from you, Dr. Anteater. Could you tell us more about ant colonies, from a reductionistic point of view?

Anteater: Gladly. As Mr. Crab mentioned to you, my profession has led me quite a long way into the understanding of ant colonies.

Achilles: I can imagine! The profession of anteater would seem to be synonymous with being an expert on ant colonies!

Anteater: I beg your pardon. "Anteater" is not my profession; it is my species. By profession, I am a colony surgeon. I specialize in correcting nervous disorders of the colony by the technique of surgical removal.

Achilles: Oh, I see. But what do you mean by "nervous disorders" of an ant colony?

Anteater: Most of my clients suffer from some sort of speech impairment. You know, colonies which have to grope for words in everyday situations. It can be quite tragic. I attempt to remedy the situation by, uhh—removing the defective part of the colony. These operations

are sometimes quite involved, and of course years of study are required before one can perform them.

Achilles: But-isn't it true that, before one can suffer from speech impairment, one must have the faculty of speech?

Anteater: Right.

Achilles: Since ant colonies don't have that faculty, I am a little confused. Crab: It's too bad, Achilles, that you weren't here last week, when Dr.

Anteater and Aunt Hillary were my house guests. I should have thought of having you over then.

Achilles: Is Aunt Hillary your aunt, Mr. Crab? Crab: Oh, no, she's not really anybody's aunt.

Anteater: But the poor dear insists that everybody should call her that, even strangers. It's just one of her many endearing quirks.

Crab: Yes, Aunt Hillary is quite eccentric, but such a merry old soul. It's a shame I didn't have you over to meet her last week.

Anteater: She's certainly one of the best-educated ant colonies I have ever had the good fortune to know. The two of us have spent many a long evening in conversation on the widest range of topics.

Achilles: I thought anteaters were devourers of ants, not patrons of antintellectualism!

Anteater: Well, of course the two are not mutually inconsistent. I am on the best of terms with ant colonies. It's just ANTS that I eat, not colonies-and that is good for both parties: me, and the colony.

Achilles: How is it possible that--

Tortoise: How is it possible that--

Achilles: -having its ants eaten can do an ant colony any good? Crab: How is it possible that

Tortoise: -having a forest fire can do a forest any good? Anteater: How is it possible that

Crab: -having its branches pruned can do a tree any good? Anteater: -having a haircut can do Achilles any good?

Tortoise: Probably the rest of you were too engrossed in the discussion to notice the lovely stretto which just occurred in this Bach fugue.

Achilles: What is a stretto?

Tortoise: Oh, I'm sorry; I thought you knew the term. It is where one theme repeatedly enters in one voice after another, with very little delay between entries.

Achilles: If I listen to enough fugues, soon I'll know all of these things and will be able to pick them out myself, without their having to be pointed out.

Tortoise: Pardon me, my friends. I am sorry to have interrupted. Dr. Anteater was trying to explain how eating ants is perfectly consistent with being a friend of an ant colony.

Achilles: Well, I can vaguely see how it might be possible for a limited and regulated amount of ant consumption to improve the overall health of

a colony-but what is far more perplexing is all this talk about having conversations with ant colonies. That's impossible. An ant colony is simply a bunch of individual ants running around at random looking for food and making a nest.

Anteater: You could put it that way if you want to insist on seeing the trees but missing the forest, Achilles. In fact, ant colonies, seen as wholes, are quite well-defined units, with their own qualities, at times including the mastery of language.

Achilles: I find it hard to imagine myself shouting something out loud in the middle of the forest, and hearing an ant colony answer back.

Anteater: Silly fellow! That's not the way it happens. Ant colonies don't converse out loud, but in writing. You know how ants form trails leading them hither and thither?

Achilles: Oh, yes-usually straight through the kitchen sink and into my peach jam.

Anteater: Actually, some trails contain information in coded form. If you know the system, you can read what they're saying just like a book. Achilles: Remarkable. And can you communicate back to them? Anteater: Without any trouble at all. That's how Aunt Hillary and I have conversations for hours. I take a stick and draw trails in the moist ground, and watch the ants follow my trails. Presently, a new trail starts getting formed somewhere. I greatly enjoy watching trails develop. As they are forming, I anticipate how they will continue (and more often I am wrong than right). When the trail is completed, I know what Aunt Hillary is thinking, and I in turn make my reply.

Achilles: There must be some amazingly smart ants in that colony, I'll say that.

Anteater: I think you are still having some difficulty realizing the difference in levels here. Just as you would never confuse an individual tree with a forest, so here you must not take an ant for the colony. You see, all the ants in Aunt Hillary are as dumb as can be. They couldn't converse to save their little thoraxes!

Achilles: Well then, where does the ability to converse come from? It must reside somewhere inside the colony! I don't understand how the ants can all be unintelligent, if Aunt Hillary can entertain you for hours with witty banter.

Tortoise: It seems to me that the situation is not unlike the composition of a human brain out of neurons. Certainly no one would insist that individual brain cells have to be intelligent beings on their own, in order to explain the fact that a person can have an intelligent conversation.

Achilles: Oh, no, clearly not. With brain cells, I see your point completely. Only ... ants are a horse of another color. I mean, ants just roam about at will, completely randomly, chancing now and then upon a morsel of food ... They are free to do what they want to do, and with that freedom, I don’t see at all how their behaviour, seen as a whole, can

amount to anything coherent-especially something so coherent as the brain behavior necessary for conversing.

Crab: It seems to me that the ants are free only within certain constraints. For example, they are free to wander, to brush against each other, to pick up small items, to work on trails, and so on. But they never step out of that small world, that ant-system, which they are in. It would never occur to them, for they don't have the mentality to imagine anything of the kind. Thus the ants are very reliable components, in the sense that you can depend on them to perform certain kinds of tasks in certain ways.

Achilles: But even so, within those limits they are still free, and they just act at random, running about incoherently without any regard for the thought mechanisms of a higher-level being which Dr. Anteater asserts they are merely components of.

Anteater: Ah, but you fail to recognize one thing. Achilles-the regularity of statistics.

Achilles: How is that?

Anteater: For example, even though ants as individuals wander about in what seems a random way, there are nevertheless overall trends, involving large numbers of ants, which can emerge from that chaos.

Achilles: Oh, I know what you mean. In fact, ant trails are a perfect example of such a phenomenon. There, you have really quite unpredictable motion on the part of any single ant-and yet, the trail itself seems to remain well-defined and stable. Certainly that must mean that the individual ants are not just running about totally at random.

Anteater: Exactly, Achilles. There is some degree of communication among the ants, just enough to keep them from wandering off completely at random. By this minimal communication they can remind each other that they are not alone but are cooperating with teammates. It takes a large number of ants, all reinforcing each other this way, to sustain any activity-such as trail-building-for any length of time. Now my very hazy understanding of the operation of brains leads me to believe that something similar pertains to the firing of neurons. Isn't it true, Mr. Crab, that it takes a group of neurons firing in order to make another neuron fire?

Crab: Definitely. Take the neurons in Achilles' brain, for example. Each neuron receives signals from neurons attached to its input lines, and if the sum total of inputs at any moment exceeds a critical threshold. then that neuron will fire and send its own output pulse rushing off to other neurons, which may in turn fire-and on down the line it goes. The neural flash swoops relentlessly in its Achillean path, in shapes stranger then the dash of a gnat-hungry swallow; every twist, every turn foreordained by the neural structure in Achilles' brain, until sensory input messages interfere.

Achilles: Normally, I think that I'M in control of what I think-but the way you put it turns it all inside out, so that it sounds as though "I" am just

what comes out of all this neural structure, and natural law. It makes what I consider my SELF sound at best like a by-product of an organism governed by natural law, and at worst, an artificial notion produced by my distorted perspective. In other words, you make me feel like I don't know who or what-I am, if anything.

Tortoise: You'll come to understand much better as we go along. But Dr.

Anteater-what do you make of this similarity?

Anteater: I knew there was something parallel going on in the two very different systems. Now I understand it much better. It seems that group phenomena which have coherence-trail-building, for example-will take place only when a certain threshold number of ants get involved. If an effort is initiated, perhaps at random, by a few ants in some locale, one of two things can happen: either it will fizzle out after a brief sputtering start

Achilles: When there aren't enough ants to keep the thing rolling?

Anteater: Exactly. The other thing that can happen is that a critical mass of ants is present, and the thing will snowball, bringing more and more ants into the picture. In the latter case, a whole "team" is brought into being which works on a single project. That project might be trailmaking, or food-gathering, or it might involve nest-keeping. Despite the extreme simplicity of this scheme on a small scale, it can give rise to very complex consequences on a larger scale.

Achilles: I can grasp the general idea of order emerging from chaos, as you sketch it, but that still is a long way from the ability to converse. After all, order also emerges from chaos when molecules of a gas bounce against each other randomly-yet all that results there is an amorphous mass with but three parameters to characterize it: volume, pressure, and temperature. Now that's a far cry from the ability to understand the world, or to talk about it!

Anteater: That highlights a very interesting difference between the explanation of the behavior of an ant colony and the explanation of the behavior of gas inside a container. One can explain the behavior of the gas simply by calculating the statistical properties of the motions of its molecules. There is no need to discuss any higher elements of structure than molecules, except the full gas itself. On the other hand, in an ant colony, you can't even begin to understand the activities of the colony unless you go through several layers of structure.

Achilles: I see what you mean. In a gas, one jump takes you from the lowest level-molecules-to the highest level-the full gas. There are no intermediate levels of organization. Now how do intermediate levels of organized activity arise in an ant colony?

Anteater: It has to do with the existence of several different varieties of ants inside any colony.

Achilles: Oh, yes. I think I have heard about that. They are called "castes", aren't they?

Anteater: That's correct. Aside from the queen, there are males, who do

practically nothing towards, the upkeep of the nest, and then—

Achilles: And of course there are soldiers-Glorious Fighters Against Communism!

Crab: Hmm ... I hardly think that could be right, Achilles. An ant colony is quite communistic internally, so why would its soldiers fight against communism? Or am I right, Dr. Anteater? .

Anteater: Yes, about colonies you are right, Mr. Crab; they are indeed based on somewhat communistic principles. But about soldiers Achilles is somewhat naive. In fact, the so-called "soldiers" are hardly adept at fighting at all. They are slow, ungainly ants with giant heads, who can snap with their strong jaws, but are hardly to be glorified. As in a true communistic state, it is rather the workers who are to be glorified. It is they who do most of the chores, such as food-gathering, hunting, and nursing of the young. It is even they who do most of the fighting.

Achilles: Bah. That is an absurd state of affairs. Soldiers who won't fight!

Anteater: Well, as I just said, they really aren't soldiers at all. It's the workers who are soldiers; the soldiers are just lazy fatheads.

Achilles: Oh, how disgraceful! Why, if I were an ant, I'd put some discipline in their ranks! I'd knock some sense into those fatheads!

Tortoise: If you were an ant? How could you be an ant? There is no way to map your brain onto an ant brain, so it seems to me to be a pretty fruitless question to worry over. More reasonable would be the proposition of mapping your brain onto an ant colony ... But let us not get sidetracked. Let Dr. Anteater continue with his most illuminating description of castes and their role in the higher levels of organization.

Anteater: Very well. There are all sorts of tasks which must be accomplished in a colony, and individual ants develop specializations. Usually an ant's specialization changes as the ant ages. And of course it is also dependent on the ant's caste. At any one moment, in any small area of a colony, there are ants of all types present. Of course, one caste may be be very sparse in some places and very dense in others.

Crab: Is the density of a given caste, or specialization, just a random thing? Or is there a reason why ants of one type might be more heavily concentrated in certain areas, and less heavily in others?

Anteater: I'm glad you brought that up, since it is of crucial importance in understanding how a colony thinks. In fact, there evolves, over a long period of time, a very delicate distribution of castes inside a colony. And it is this distribution which allows the colony to have the complexity which underlies the ability to converse with me.

Achilles: It would seem to me that the constant motion of ants to and fro would completely prevent the possibility of a very delicate distribution.Any delicate distribution would be quickly destroyed by all the random motions of ants, just as any delicate pattern among molecules in a gas would not survive for an instant, due to the random bombardment from all sides.

Anteater: In an ant colony. the situation is quite the contrary. In fact, it is just exactly the to-ing and fro-ing of ants inside the colony

which adapts the caste distribution to varying situations, and thereby preserves the delicate caste distribution. You see, the caste distribution cannot remain as one single rigid pattern; rather, it must constantly be changing so as to reflect, in some manner, the real-world situation with which the colony is dealing, and it is precisely the motion inside the colony which updates the caste distribution, so as to keep it in line with the present circumstances facing the colony.

Tortoise: Could you give an example?

Anteater: Gladly. When I, an anteater, arrive to pay a visit to Aunt Hillary, all the foolish ants, upon sniffing my odor, go into a panic-which means, of course, that they begin running around completely differently from the way they were before I arrived.

Achilles: But that's understandable, since you're a dreaded enemy of the colony.

Anteater: Oh, no. I must reiterate that, far from being an enemy of the colony, I am Aunt Hillary's favorite companion. And Aunt Hillary is my favorite aunt. I grant you, I'm quite feared by all the individual ants in the colony-but that's another matter entirely. In any case, you see that the ants' action in response to my arrival completely changes the internal distribution of ants.

Achilles: That's clear.

Anteater: And that sort of thing is the updating which I spoke of. The new distribution reflects my presence. One can describe the change from old state to new as having added a "piece of knowledge" to the colony.

Achilles: How can you refer to the distribution of different types of ants inside a colony as a "piece of knowledge"?

Anteater: Now there's a vital point. It requires some elaboration. You see, what it comes down to is how you choose to describe the caste distribution. If you continue to think in terms of the lower levels-individual ants-then you miss the forest for the trees. That's just too microscopic a level, and when you think microscopically, you're bound to miss some large-scale features. You've got to find the proper high-level framework in which to describe the caste distribution-only then will it make sense how the caste distribution can encode many pieces of knowledge.

Achilles: Well, how DO you find the proper-sized units in which to describe the present state of the colony, then?

Anteater: All right. Let's begin at the bottom. When ants need to get something done, they form little "teams", which stick together to perform a chore. As I mentioned earlier, small groups of ants are constantly forming and unforming. Those which actually exist for a while are the teams, and the reason they don't fall apart is that there really is something for them to do.

Achilles: Earlier you said that a group will stick together if its size exceeds a certain threshold. Now you're saying that a group will stick together if there is something for it to do.

Anteater: They are equivalent statements. For instance, in food-gathering,

if there is an inconsequential amount of food somewhere which gets discovered by some wandering Ant who then attempts to communicate its enthusiasm to other ants, the number of ants who respond will be proportional to the size of the food sample-and an inconsequential amount will not attract enough ants to surpass the threshold. Which is exactly what I meant by saying there is nothing to do-too little food ought to be ignored.

Achilles: I see. I assume that these "teams" are one of the levels of structure falling somewhere in between the single-ant level and the colony level.

Anteater: Precisely. There exists a special kind of team, which I call a "signal"-and all the higher levels of structure are based on signals. In fact, all the higher entities are collections of signals acting in concert. There are teams on higher levels whose members are not ants, but teams on lower levels. Eventually you reach the lowest-level teams which is to say, signals-and below them, ants.

Achilles: Why do signals deserve their suggestive name?

Anteater: It comes from their function. The effect of signals is to transport ants of various specializations to appropriate parts of the colony. So the typical story of a signal is thus: it comes into existence by exceeding the threshold needed for survival, then it migrates for some distance through the colony, and at some point it more or less disintegrates into its individual members, leaving them on their own.

Achilles: It sounds like a wave, carrying sand dollars and seaweed from afar, and leaving them strewn, high and dry, on the shore.

4nteater: In a way that's analogous, since the team does indeed deposit something which it has carried from a distance, but whereas the water in the wave rolls back to the sea, there is no analogous carrier substance in the case of a signal, since the ants themselves compose it.

Tortoise: And I suppose that a signal loses its coherency just at some spot in the colony where ants of that type were needed in the first place.

Anteater: Naturally.

Achilles: Naturally? It's not so obvious to ME that a signal should always go just where it is needed. And even if it goes in the right direction, how does it figure out where to decompose? How does it know it has arrived?

Anteater: Those are extremely important matters, since they involve explaining the existence of purposeful behavior-or what seems to be purposeful behavior-on the part of signals. From the description, one would be inclined to characterize the signals' behavior as being oriented towards filling a need, and to call it "purposeful". But you can look at it otherwise.

Achilles: Oh, wait. Either the behavior is purposeful, or it is NOT. I don't see how you can have it both ways.

Anteater: Let me explain my way of seeing things, and then see if you agree. Once a signal is formed, there is no awareness on its part that it

should head off in any particular direction. But here, the delicate caste distribution plays a crucial role. It is what determines the motion of signals through the colony, and also how long a signal will remain stable, and where it will "dissolve".

Achilles: So everything depends on the caste distribution, eh?

Anteater: Right. Let's say a signal is moving along. As it goes, the ants which compose it interact, either by direct contact or by exchange of scents, with ants of the local neighborhoods which it passes through. The contacts and scents provide information about local matters of urgency, such as nest-building, or nursing, or whatever. The signal will remain glued together as long as the local needs are different from what it can supply; but if it CAN contribute, it disintegrates, spilling a fresh team of usable ants onto the scene. Do you see now how the caste distribution acts as an overall guide of the teams inside the colony?

Achilles: I do see that.

Anteater: And do you see how this way of looking at things requires attributing no sense of purpose to the signal?

Achilles: I think so. Actually, I'm beginning to see things from two different vantage points. From an ant's-eye point of view, a signal has NO purpose. The typical ant in a signal is just meandering around the colony, in search of nothing in particular, until it finds that it feels like stopping. Its teammates usually agree, and at that moment the team unloads itself by crumbling apart, leaving just its members but none of its coherency. No planning is required, no looking ahead; nor is any search required, to determine the proper direction. But from the COLONY'S point-of view, the team has just responded to a message which was written in the language of the caste distribution. Now from this perspective, it looks very much like purposeful activity.

Crab: What would happen if the caste distribution were entirely random? Would signals still band and disband?

Anteater: Certainly. But the colony would not last long, due to the meaninglessness of the caste distribution.

Crab: -Precisely the point I wanted to make. Colonies survive because their caste distribution has meaning, and that meaning is a holistic aspect, invisible on lower levels. You lose explanatory power unless you take that higher level into account.

Anteater: I see your side; but I believe you see things too narrowly.

Crab: How so?

Anteater: Ant colonies have been subjected to the rigors of evolution for billions of years. A few mechanisms were selected for, and most were selected against. The end result was a set of mechanisms which make ant colonies work as we have been describing. If you could watch the whole process in a movie-running a billion or so times faster than life, of course-the emergence of various mechanisms would be seen as natural responses to external pressures, just as bubbles in boiling water are natural responses to an external heat source. I don't suppose you

see "meaning" and "purpose", in the bubbles in boiling water-or do you?

Crab: No, but

Anteater: Now that's MY point. No matter how big a bubble is, it owes its existence to processes on the molecular level, and you can forget about any "higher-level laws". The same goes for ant colonies and their teams. By looking at things from the vast perspective of evolution, you can drain the whole colony of meaning and purpose. They become superfluous notions.

Achilles: Why, then, Dr. Anteater, did you tell me that you talked with Aunt Hillary? It now seems that you would deny that she can talk or think at all.

Anteater: I am not being inconsistent, Achilles. You see, I have as much difficulty as anyone else in seeing things on such a grandiose time scale, so I find it much easier to change points of view. When I do so, forgetting about evolution and seeing things in the here and now, the vocabulary of teleology comes back: the MEANING of the caste distribution and the PURPOSEFULNESS of signals. This not only happens when I think of ant colonies, but also when I think about my own brain and other brains. However, with some effort I can always remember the other point of view if necessary, and drain all these systems of meaning, too.

Crab: Evolution certainly works some miracles. You never know the next trick it will pull out of its sleeve. For instance, it wouldn't surprise me one bit if it were theoretically possible for two or more "signals" to pass through each other, each one unaware that the other one is also a signal; each one treating the other as if it were just part of the background population.

Anteater: It is better than theoretically possible; in fact it happens routinely!

Achilles: Hmm ... What a strange image that conjures up in my mind. I can just imagine ants moving in four different directions, some black, some white, criss-crossing, together forming an orderly pattern, almost like-like

Tortoise: A fugue, perhaps?

Achilles: Yes-that's it! An ant fugue!

Crab: An interesting image, Achilles. By the way, all that talk of boiling water made me think of tea. Who would like some more? Achilles: I could do with another cup, Mr. C.

Crab: Very good.

Achilles: Do you suppose one could separate out the different visual "voices" of such an "ant fugue"? I know how hard it is for me

Tortoise: Not for me, thank you.

Achilles: -to track a single voice

Anteater: I'd like some, too, Mr. Crab

Achilles: -- in a musical fugue--

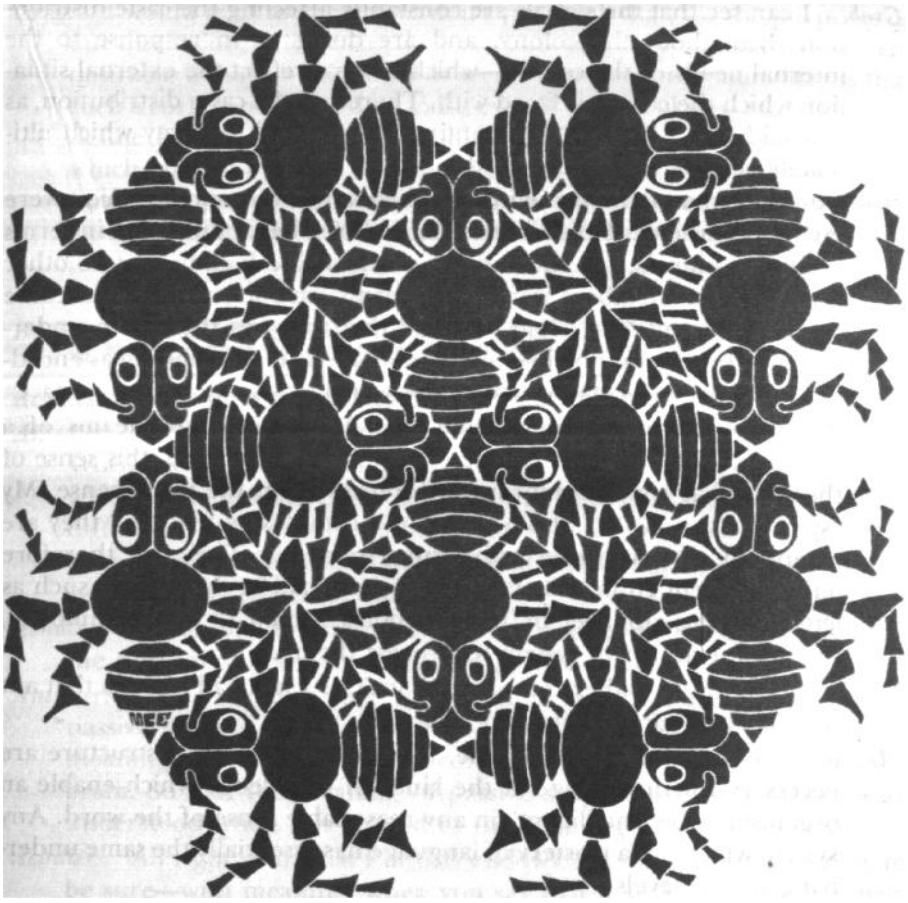


FIGURE 61. "Ant Fugue", by M. C. Escher (woodcut, 1953).

Anteater: -if it isn't too much trouble
Achilles: . -when all of them
Crab: Not at all. Four cups of tea
Tortoise: Three?
Achilles: -are going at once.
Crab: -coming right up!
Anteater: That's an interesting thought, Achilles. But its unlikely that anyone could draw such a picture in a convincing way
Achilles: That's too bad.
Tortoise: Perhaps you could answer this, Dr. Anteater. Does a signal, from its creation until its dissolution, always consist of the same set of ants?
Anteater: As a matter of fact, the individuals in a signal sometimes break off and get replaced by others of the same caste, if there are a few in the area. Most often, signals arrive at their disintegration points with nary an ant in common with their starting lineup.

Crab: I can see that the signals are constantly affecting the caste distribution throughout the colony, and are doing so in response to the internal needs of the colony-which in turn reflect the external situation which the colony is faced with. Therefore the caste distribution, as you said, Dr. Anteater, gets continually updated in a way which ultimately reflects the outer world.

Achilles: But what about those intermediate levels of structure? You were saying that the caste distribution should best be pictured not in terms of ants or signals, but in terms of teams whose members were other teams, whose members were other teams, and so on until you come down to the ant level. And you said that that was the key to understanding how it was possible to describe the caste distribution as encoding pieces of information about the world.

Anteater: Yes, we are coming to all that. I prefer to give teams of a sufficiently high level the name of "symbols". Mind you, this sense of the word has some significant differences from the usual sense. My "symbols" are ACTIVE SUBSYSTEMS of a complex system, and they are composed of lower-level active subsystems ... They are therefore quite different from PASSIVE symbols, external to the system, such as letters of the alphabet or musical notes, which sit there immobile, waiting for an active system to process them.

Achilles: Oh, this is rather complicated, isn't it? I just had no idea that ant colonies had such an abstract structure.

Anteater: Yes, it's quite remarkable. But all these layers of structure are necessary for the storage of the kinds of knowledge which enable an organism to be "intelligent" in any reasonable sense of the word. Any system which has a mastery of language has essentially the same underlying sets of levels.

Achilles: Now just a cotton-picking minute. Are you insinuating that my brain consists of, at bottom, just a bunch of ants running around?

Anteater: Oh, hardly. You took me a little too literally. The lowest level may be utterly different. Indeed, the brains of anteaters, for instance, are not composed of ants. But when you go up a level or two in a brain, you reach a level whose elements have exact counterparts in other systems of equal intellectual strength-such as ant colonies.

Tortoise: That is why it would be reasonable to think of mapping your brain, Achilles, onto an ant colony, but not onto the brain of a mere ant.

Achilles: I appreciate the compliment. But how would such a mapping be carried out? For instance, what in my brain corresponds to the low level teams which you call signals?

Anteater: Oh, I but dabble in brains, and therefore couldn't set up the map in its glorious detail. But-and correct me if I'm wrong, Mr. Crab-I would surmise that the brain counterpart to an ant colony's signal is the firing of a neuron; or perhaps it is a larger-scale event, such as a pattern of neural firings.

Crab: I would tend to agree. But don't you think that, for the purposes of our discussion, delineating the exact counterpart is not in itself crucial, desirable though it might be? It seems to me that the main idea is that such a correspondence does exist, even if we don't know exactly how to define it right now. I would only question one point, Dr. Anteater, which you raised, and that concerns the level at which one can have faith that the correspondence begins. You seemed to think that a SIGNAL might have a direct counterpart in a brain; whereas I feel that it is only at the level of your ACTIVE SYMBOLS and above that it is likely that a correspondence must exist.

Anteater: Your interpretation may very well be more accurate than mine, Mr. Crab. Thank you for bringing out that subtle point.

Achilles: What does a symbol do that a signal couldn't do?

Anteater: It is something like the difference between words and letters. Words, which are meaning-carrying entities, are composed of letters, which in themselves carry no meaning. This gives a good idea of the difference between symbols and signals. In fact it is a useful analogy, as long as you keep in mind the fact that words and letters are PASSIVE, symbols and signals are ACTIVE.

Achilles: I'll do so, but I'm not sure I understand why it is so vital to stress the difference between active and passive entities.

Anteater: The reason is that the meaning which you attribute to any passive symbol, such as a word on a page, actually derives from the meaning which is carried by corresponding active symbols in your brain. So that the meaning of passive symbols can only be properly understood when it is related to the meaning of active symbols.

Achilles: All right. But what is it that endows a SYMBOL-an active one, to be sure-with meaning, when you say that a SIGNAL, which is a perfectly good entity in its own right, has none? Anteater: It all has to do with the way that symbols can cause other symbols to be triggered. When one symbol becomes active, it does not do so in isolation. It is floating about, indeed, in a medium, which is characterized by its caste distribution.

Crab: Of course, in a brain there is no such thing as a caste distribution, but the counterpart is the "brain state". There, you describe the states of all the neurons, and all the interconnections, and the threshold for firing of each neuron.

Anteater: Very well; let's lump "caste distribution" and "brain state" under a common heading, and call them just the "state". Now the state can be described on a low level or on a high level. A low-level description of the state of an ant colony would involve painfully specifying the location of each ant, its age and caste, and other similar items. A very detailed description, yielding practically no global insight as to WHY it is in that state. On the other hand, a description on a high level would involve specifying which symbols could be triggered by which combinations of other symbols, under what conditions, and so forth.

Achilles: What about a description on the level of signals, or teams?

Anteater: A description on that level would fall somewhere in between the low-level and symbol-level descriptions. It would contain a great deal of information about what is actually going on in specific locations throughout the colony, although certainly less than an ant-by-ant description, since teams consist of clumps of ants. A team-by-team description is like a summary of an ant-by-ant description. However, you have to add extra things which were not present in the ant-by-ant description-such as the relationships between teams, and the supply of various castes here and there. This extra complication is the price you pay for the right to summarize.

Achilles: It is interesting to me to compare the merits of the descriptions at various levels. The highest-level description seems to carry the most explanatory power, in that it gives you the most intuitive picture of the ant colony, although strangely enough, it leaves out seemingly- the most important feature-the ants.

Anteater: But you see, despite appearances, the ants are not the most important feature. Admittedly, were it not for them, the colony Wouldn't exist: but something equivalent-a brain-can exist, ant-free. So, at least from a high-level point of view, the ants are dispensable. .Achilles: I'm sure no ant would embrace your theory with eagerness.

Anteater: Well, I never met an ant with a high-level point of view.

Crab: What a counterintuitive picture you paint, Dr. Anteater. It seems that, if what you say is true, in order to grasp the whole structure, you have to describe it omitting any mention of its fundamental building blocks.

Anteater: Perhaps I can make it a little clearer by an analogy. Imagine you have before you a Charles Dickens novel.

Achilles: The Pickwick Papers-will that do?

Anteater: Excellently! And now imagine trying the following game: you must find a way of mapping letters onto ideas, so that the entire Pickwick Papers makes sense when you read it letter by letter.

Achilles: Hmm ... You mean that every time I hit a word such as "the", I have to think of three definite concepts, one after another, with no room for variation?

Anteater: Exactly. They are the `t'-concept, the `h'-concept, and the `e'-concept-and every time, those concepts are as they were the preceding time.

Achilles: Well, it sounds like that would turn the experience of "reading" The Pickwick Papers into an indescribably boring nightmare. It would be an exercise in meaninglessness, no matter what concept I associated with each letter.

Anteater: Exactly. There is no natural mapping from the individual letters into the real world. The natural mapping occurs on a higher level between words, and parts of the real world. If you wanted to describe the book, therefore, you would make no mention of the letter level.

Achilles: Of course not! I'd describe the plot and the characters, and so forth.

Anteater: So there you are. You would omit all mention of the building blocks, even though the book exists thanks to them. They are the medium, but not the message.

Achilles: All right-but what about ant colonies?

Anteater: Here, there are active signals instead of passive letters, and active symbols instead of passive words-hut the idea carries over.

Achilles: Do you mean I couldn't establish a mapping between signals and things in the real world?

Anteater: You would find that you could not do it in such a way that the triggering of new signals would make am sense. Nor could you succeed on any lower level-for example the ant level. Only on the symbol level do the triggering patterns make sense. Imagine, for instance, that one day you were watching Aunt Hillary when I arrived to pay a call. You could watch as carefully as you wanted, and yet you would probably perceive nothing more than a rearrangement of ants.

Achilles: I'm sure that's accurate.

Anteater: And yet, as I watched, reading the higher level instead of the lower level, I would see several dormant symbols being awakened, those which translate into the thought, "Oh, here's that charming Dr. Anteater again-how pleasant!"-or words to that effect.

Achilles: That sounds like what happened when the four of us all found different levels to read in the MU-picture--or at least THREE of us did .. .

Tortoise: What an astonishing coincidence that there should be such a resemblance between that strange picture which I chanced upon in the Well-Tempered Clavier, and the trend of our conversation.

Achilles: Do you think it's just coincidence?

Tortoise: Of course.

Anteater: Well, I hope you can grasp now how the thoughts in Aunt Hillary emerge from the manipulation of symbols composed of signals composed of teams composed of lower-level teams, all the way down to ants.

Achilles: Why do you call it "symbol manipulation"? Who does the manipulating, if the symbols are themselves active? Who is the agent?

Anteater: This gets back to the question which you earlier raised about purpose. You're right that symbols themselves are active, but the activities which they follow are nevertheless not absolutely free. The activities of all symbols are strictly determined by the state of the full system in which they reside. Therefore, the full system is responsible for how its symbols trigger each other, and so it is quite reasonable to speak of the full system as the "agent". As the symbols operate, the state of the system gets slowly transformed, or updated. But there are many features which remain over time. It is this partially constant, partially varying system which is the agent. One can give a name to the

full system. For example, Aunt Hillary is the "who" who can be said to manipulate her symbols; and you are similar, Achilles.

Achilles: That's quite a strange characterization of the notion of who I am. I'm not sure I can fully understand it, but I will give it some thought.

Tortoise: It would be quite interesting to follow the symbols in your brain as you do that thinking about the symbols in your brain.

Achilles: That's too complicated for me. I have trouble enough just trying to picture how it is possible to look at an ant colony and read it on the symbol level. I can certainly imagine perceiving it at the ant level; and with a little trouble, I can imagine what it must be like to perceive it at the signal level; but what in the world can it be like to perceive an ant colony at the symbol level?

Anteater: One only learns through long practice. But when one is at my stage, one reads the top level of an ant colony as easily as you yourself read the "MU" in the MU-picture.

Achilles: Really? That must be an amazing experience.

Anteater: In a way-but it is also one which is quite familiar to you, Achilles.

Achilles: Familiar to me? What do you mean? I have never looked at an ant colony on anything but the ant level.

Anteater: Maybe not; but ant colonies are no different from brains in many respects.

Achilles: I have never seen nor read any brain either, however.

Anteater: What about your OWN brain? Aren't you aware of your own thoughts? Isn't that the essence of consciousness? What else are you doing but reading your own brain directly at the symbol level?

Achilles: I never thought of it that way. You mean that I bypass all the lower levels, and only see the topmost level?

Anteater: That's the way it is, with conscious systems. They perceive themselves on the symbol level only, and have no awareness of the lower levels, such as the signal levels.

Achilles: Does it follow that in a brain, there are active symbols which are constantly updating themselves so that they reflect the overall state of the brain itself, always on the symbol level?

Anteater: Certainly. In any conscious system there are symbols which represent the brain state, and they are themselves part of the very brain state which they symbolize. For consciousness requires a large degree of self-consciousness.

Achilles: That is a weird notion. It means that although there is frantic activity occurring in my brain at all times, I am only capable of registering that activity in one way-on the symbol level; and I am completely insensitive to the lower levels. It is like being able to read a Dickens novel by direct visual perception, without ever having learned the letters of the alphabet. I can't imagine anything as weird as that really happening.

Crab: But precisely that sort of thing can happen when you read “MU”,

without perceiving the lower levels "HOLISM" and "REDUCTIONISM".

Achilles: You're right-I bypassed the lower levels, and saw only the top. I wonder if I'm missing all sorts of meaning on lower levels of my brain as well, by reading only the symbol level. It's too bad that the top level doesn't contain all the information about the bottom level, so that by reading the top, one also learns what the bottom level says. But I guess it would be naive to hope that the top level encodes anything from the bottom level-it probably doesn't percolate up. The MU-picture is the most striking possible example of that: there, the topmost level says only "ML which bears no relation whatever to the lower levels!

Crab: That's absolutely true. (*Picks up the MU-picture, to inspect it more closely*.) Hmm ... There's something strange about the smallest letters in this picture; they're very wiggly ... '

Anteater: Let me take a look. (*Peers closely at the MU-picture*.) I think there's yet another level, which all of us missed!

Tortoise: Speak for yourself, Dr. Anteater.

Achilles: Oh, no-that can't be! Let me see. (*Looks very carefully*.) I know the rest of you won't believe this, but the message of this picture is staring us all in the face, hidden in its depths. It is simply one word, repeated over and over again, like a mantra-but what an important one: "Mu"! What do you know! It is the same as the top level! And none of us suspected it in the least.

Crab: We would never have noticed it if it hadn't been for you, Achilles. Anteater: I wonder if the coincidence of the highest and lowest levels happened by chance? Or was it a purposeful act carried out by some creator?

Crab: How could one ever decide that?

Tortoise: I don't see any way to do so, since we have no idea why that particular picture is in the Crab's edition of the Well-Tempered Clavier. Anteater: Although we have been having a lively discussion, I have still managed to listen with a good fraction of an ear to this very long and complex four-voice fugue. It is extraordinarily beautiful.

Tortoise: It certainly is. And now, in just a moment, comes an organ point.

Achilles: Isn't an organ point what happens when a piece of music slows down slightly, settles for a moment or two on a single note or chord, and then resumes at normal speed after a short silence?

Tortoise: No, you're thinking of a "fermata"-a sort of musical semicolon. Did you notice there was one of those in the prelude?

Achilles: I guess I must have missed it.

Tortoise: Well, you have another chance coming up to hear a fermata-in fact, there are a couple of them coming up, towards the end of this fugue.

Achilles: Oh, good. You'll point them out in advance, won't you? Tortoise: If you like.

Achilles: But do tell me, what is an organ point?

Tortoise: An organ point is the sustaining of a single note by one of the

voices in a polyphonic piece (often the lowest voice), while the other voices continue their own independent lines, This organ point is on the note of G. Listen carefully, and you'll hear it.

Anteater:. There occurred an incident one day when I visited with Aunt Hillary which reminds me of your suggestion of observing the symbols in Achilles' brain as they create thoughts which are about themselves.

Crab: Do tell us about it.

Anteater: Aunt Hillary had been feeling very lonely, and was very happy to have someone to talk to that day. So she gratefully told me to help myself to the juiciest ants I could find. (She's always been most generous with her ants.)

Achilles: Gee!

Anteater: It just happened that I had been watching the symbols which were carrying out her thoughts, because in them were some particularly juicy-looking ants.

Achilles: Gee!

Anteater: So I helped myself to a few of the fattest ants which had been parts of the higher-level symbols which I had been reading. Specifically, the symbols which they were part of were the ones which had expressed the thought, "Help yourself to any of the ants which look appetizing."

Achilles: Gee!

Anteater: Unfortunately for them, but fortunately for me, the little bugs didn't have the slightest inkling of what they were collectively telling me, on the symbol level.

Achilles: Gee! That is an amazing wraparound. They were completely unconscious of what they were participating in. Their acts could be seen as part of a pattern on a higher level, but of course they were completely unaware of that. Ah, what a pity-a supreme irony, in fact-that they missed it.

Crab: You are right, Mr. T-that was a lovely organ point.

Anteater: I had never heard one before, but that one was so conspicuous that no one could miss it. Very effective.

Achilles: What? Has the organ point already occurred? How can I not have noticed it, if it was so blatant?

Tortoise: Perhaps you were so wrapped up in what you were saying that you were completely unaware of it. Ah, what a pity-a supreme irony, in fact-that you missed it.

Crab: Tell me, does Aunt Hillary live in an anthill?

Anteater: Well, she owns a rather large piece of property. It used to belong to someone else, but that is rather a sad story. In any case, her estate is quite expansive. She lives rather sumptuously, compared to many other colonies.

!chilies: How does that jibe with the communistic nature of ant colonies which you earlier described to us? It sounds quite inconsistent, to me, to preach communism and to live in a fancy estate.

Anteater: The communism is on the ant level. In an ant colony all ants work for the common good, even to their own individual detriment at times. Now this is simply a built-in aspect of Aunt Hillary's structure, but for all I know, she may not even be aware of this internal communism. Most human beings are not aware of anything about their neurons; in fact they probably are quite content not to know anything about their brains, being somewhat squeamish creatures. Aunt Hillary is also somewhat squeamish; she gets rather antsy whenever she starts to think about ants at all. So she avoids thinking about them whenever possible. I truly doubt that she knows anything about the communistic society which is built into her very structure. She herself is a staunch believer in libertarianism-you know, laissez-faire and all that. So it makes perfect sense, to me at least, that she should live in a rather sumptuous manor.

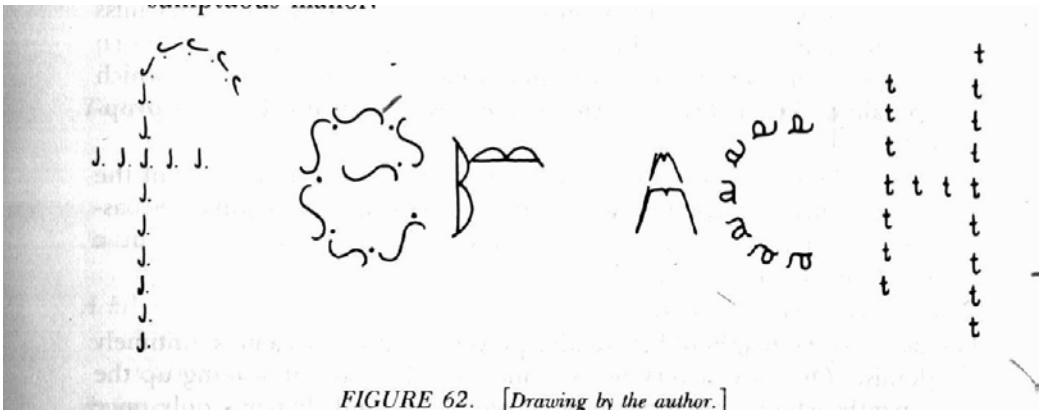


FIGURE 62. [*Drawing by the author.*]

Tortoise: As I turned the page just now, while following along in this lovely edition of the Well-Tempered Clavier, I noticed that the first of the two fermatas is coming up soon-so you might listen for it, Achilles. Achilles: I will, I will.

Tortoise: Also, there's a most curious picture facing this page. Crab: Another one? What next?

Tortoise: See for yourself. (Passes the score over to the Crab.)

Crab: Aha! It's just a few bunches of letters. Let's see-there are various numbers of the letters `J', 'S', `B', `m', `a', and 't'. It's strange, how the first three letters grow, and then the last three letters shrink again. Anteater: May I see it?

Crab: Why, certainly.

Anteater: Oh, by concentrating on details, you have utterly missed the big picture. In reality, this group of letters is `f', `e', `r', 'A', `C', 'H', without any repetitions. First they get smaller, then they get bigger. Here, Achilles-what do you make of it?

Achilles: Let me see. Hmm. Well, I see it as a set of upper-case letters which grow as you move to the right.

Tortoise: Do they spell anything?

Achilles: Ah ... "J. S. BACH". Oh! I understand now. It's Bach's name!

Tortoise: Strange that you should see it that way. I see it as a set of lower-case letters, shrinking as they move to the right, and ... spelling out ... the name of ... (Slows down slightly, especialh drawing out the last few words. Then there is a brief silence. Suddenly he resumes as if nothing unusual had happened.) -"fermat".

Achilles: Oh, you've got Fermat on the brain, I do believe. You see Fermat's Last Theorem everywhere.

Anteater: You were right, Mr. Tortoise-I just heard a charming little fermata in the fugue.

Crab: So did I.

Achilles: Do you mean everybody heard it but me? I'm beginning to feel stupid.

Tortoise: There, there, Achilles-don't feel bad. I'm sure you won't miss Fugue's Last Fermata (which is coming up quite soon). But, to return to our previous topic, Dr. Anteater, what is the very sad story which you alluded to, concerning the former owner of Aunt Hillary's property

Anteater: The former owner was an extraordinary individual, one of the most creative ant colonies who ever lived. His name was Johant Sebastiant Fermant, and he was a mathematiciant by vocation, but a musiciant by avocation.

Achilles: How very versantile of him!

Anteater: At the height of his creative powers, he met with a most untimely demise. One day, a very hot summer day, he was out soaking up the warmth, when a freak thundershower-the kind that hits only once every hundred years or so-appeared from out of the blue, and thoroughly drenched J. S F. Since the storm came utterly without warning, the ants got completely disoriented and confused. The intricate organization which had been so finely built up over decades, all went down the drain in a matter of minutes. It was tragic.

Achilles: Do you mean that all the ants drowned, which obviously would spell the end of poor J. S. F.

Anteater: Actually, no. The ants managed to survive, every last one of them, by crawling onto various sticks and logs which floated above the raging torrents. But when the waters receded and left the ants back on their home grounds, there was no organization left. The caste distribution was utterly destroyed, and the ants themselves had no ability to reconstruct what had once before been such a finely tuned organization. They were as helpless as the pieces of Humpty Dumpty in putting themselves back together again. I myself tried, like all the king's horses and all the king's men, to put poor Fermant together again. I faithfully put out sugar and cheese, hoping against hope that somehow Fermant would reappear ... (*Pulls out a handkerchief and wipes his eyes.)*

Achilles: How valiant of you! I never knew Anteaters had such big hearts.

Anteater: But it was all to no avail. He was Bone, beyond reconstitution.

However, something very strange then began to take place: over the next few months, the ants which had been components of J. S. F. slowly regrouped, and built up a new organization. And thus was Aunt Hillary born.

Crab: Remarkable! Aunt Hillary is composed of the very same ants as Fermant was

Anteater: Well, originally she was, yes. By now, some of the older ants have died, and been replaced. But there are still many holdovers from the J. S. F.-days.

Crab: And can't you recognize some of J. S. F.'s old traits coming to the fore, from time to time, in Aunt Hillary%

Anteater: Not a one. They have nothing in common. And there is no reason they should, as I see it. There are, after all, often several distinct ways to rearrange a group of parts to form a "sum". And Aunt Hillary was just a new "sum" of the old parts. Not MORE than the sum, mind you just that particular KIND of sum.

Tortoise: Speaking of sums, I am reminded of number theory, where occasionally one will be able to take apart a theorem into its component symbols, rearrange them in a new order, and come up with a new theorem.

Anteater: I've never heard of such a phenomenon, although I confess to being a total ignoramus in the field.

Achilles: Nor have I heard of it-and I am rather well versed in the field, If I don't say so myself. I suspect Mr. T is just setting up one of his elaborate spoofs. I know him pretty well by now. Anteater: Speaking of number theory, I am reminded of J. S. F. again, for number theory is one of the domains in which he excelled. In fact, he made some rather rema, ..able contributions to number theory. Aunt Hillary, on the other hand, is remarkably dull-witted in anything that has even the remotest connection with mathematics. Also, she has only a rather banal taste in music, whereas Sebastiant was extremely gifted in music.

Achilles: I am very fond of number theory. Could you possibly relate to us something of the nature of Sebastiant's contributions,

Anteater: Very well, then. (Pauses for a moment to sip his tea, then resumes.)

Have you heard of Fourmi's infamous "Well-Tested Conjecture”?

Achilles. I'm not sure ... It sounds strangely familiar, and yet I can't quite place it.

Anteater: It's a very simple idea. Lierre de Fourmi, a mathematiciant by vocation but lawyer by avocation, had been reading in his copy-of the classic text Arithmetica by Di of Antus, and came across a page containing the equation

$$2^a+2^b=2^c$$

He immediately realized that this equation has infinitely many solutions a. b, c, and then wrote in the margin the following notorious comment:

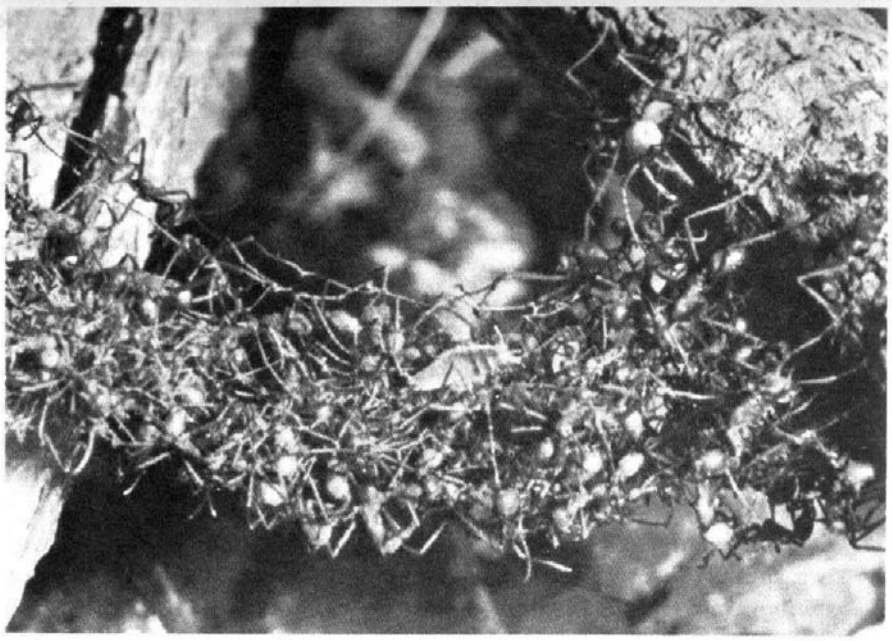


FIGURE 63. *During emigrations arm' ants sometimes create living bridges of their own bodies. In this photograph of such a bridge (de Fourmi Lierre), the workers of an Eciton burchelli colony can be seen linking their legs and, along the top of the bridge, hooking their tarsal claws together to form irregular systems of chains. .A symbiotic silverfish, Trichatelura manni, is seen crossing the bridge in the center. [From E. O. Wilson, The Insect Societies 'Cambridge, Mass.: Harvard University Press, 1971), p. 62)*

The equation

$$n^a+n^b=n^c$$

has solutions in positive integers a, b, c, and n only when n = 2 (and then there are infinitely many triplets a, b, c which satisfy the equation); but there are no solutions for n > 2. I have discovered a truly marvelous proof of this statement, which, unfortunately. is so small that it would be well-nigh invisible if written in the margin. Ever since that year, some three hundred days ago, mathematicians have been vainly trying to do one of two things: either to prove Fourmi's claim, and thereby vindicate Fourmi's reputation, which, although very high, has been somewhat tarnished by skeptics who think he never really found the proof he claimed to have found-or else to refute the claim, by finding a counterexample: a set of four integers a, b, c, and n, with n > 2, which satisfy the equation. Until very recently, every attempt in either direction had met with failure. To be sure, the Conjecture has been verified for many specific values of n-in particular, all n up to 125,000. But no one had succeeded in proving it for ALL n-no one, that is, until Johant Sebastiant Fermant came upon the scene. It was he who found the proof that cleared Fourmi's name.

It now goes under the name "Johant Sebastiant's Well-Tested Conjecture".

Achilles: Shouldn't it be called a "Theorem" rather than a "Conjecture", if it's finally been given a proper proof;

Anteater: Strictly speaking, you're right, but tradition has kept it this way.

Tortoise: What sort of music did Sebastiant do?

Anteater: He had great gifts for composition. Unfortunately, his greatest work is shrouded in mystery, for he never reached the point of publishing it. Some believe that he had it all in his mind; others are more unkind, saying that he probably never worked it out at all, but merely blustered about it.

Achilles: What was the nature of this magnum opus?

Anteater: It was to be a giant prelude and fugue; the fugue was to have twenty-four voices, and to involve twenty-four distinct subjects, one in each of the major and minor keys.

Achilles: It would certainly be hard to listen to a twenty-four-voice fugue as a whole!

Crab: Not to mention composing one!

Anteater: But all that we know of it is Sebastiant's description of it, which he wrote in the margin of his copy of Buxtehude's Preludes and Fugues for Organ. The last words which he wrote before his tragic demise were:

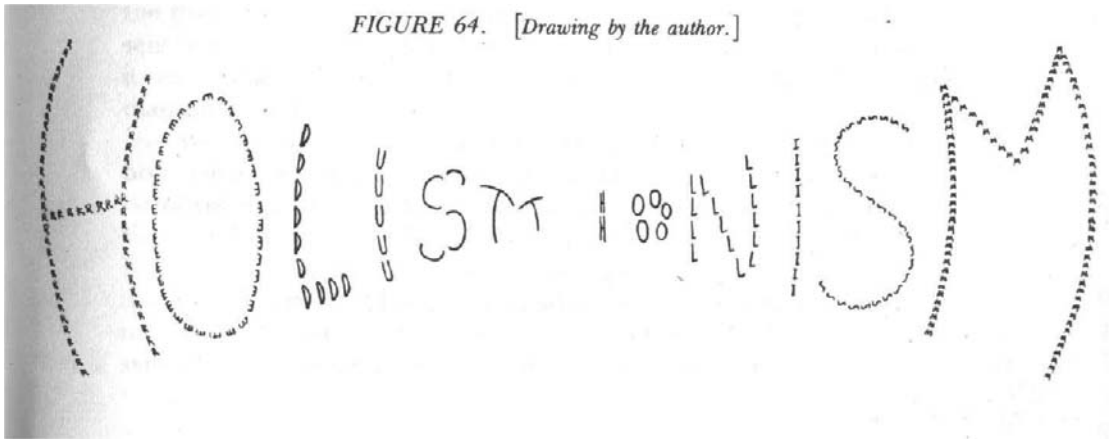
> I have composed a truly marvelous fugue. In it, I have added together the power of 24 keys, and the power of 24 themes; I came up with a fugue with the power of 24 voices. Unfortunately, this margin is too narrow to contain it.

And the unrealized masterpiece simply goes by the name, "Fermant's Last Fugue".

Achilles: Oh, that is unbearably tragic.

Tortoise: Speaking of fugues, this fugue which we have been listening to is nearly over. Towards the end, there occurs a strange new twist on its theme. (Flips the page in the Well-Tempered Clavier.) Well, what have we here? A new illustration-how appealing! *(Shows it to the Crab.)*

FIGURE 64. [*Drawing by the author.*]



Crab: Well, what have we here? Oh, I see: It´s HOLISMIONSIM", written in large letters that first shrink and then grow back to their original size. But that doesn't make any sense, because it's not a word. Oh me, oh my! (Passes it to the Anteater.)

Anteater: Well, what have we here? Oh, I see: it's "REDUCTHOLISM", written in small letters that first grow and then shrink back to their original size. But that doesn't make any sense, because it's not a word. Oh my, oh me! (Passes it to Achilles.)

Achilles: I know the rest of you won't believe this, but in fact this picture consists of the word "HOLISM" written twice, with the letters continually shrinking as they proceed from left to right. (Returns it to the Tortoise.)

Tortoise: I know the rest of you won't believe this, but in fact this picture consists of the word "REDUCTIONISM" written once, with the letters continually growing as they proceed from left to right.

Achilles: At last-I heard the new twist on the theme this time! I am so glad that you pointed it out to me, Mr. Tortoise. Finally, I think I am beginning to grasp the art of listening to fugues.